Chemistry Practice Questions — 150 Problems

> **Beginner (C1–C50): Level 1–2, 1–3 cards. Intermediate (C51–C100): Level 2–3, 2–5 cards. Advanced (C101–C150): Level 3–4, 4–10 cards.**
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> Every [Cards: …] tag is an authoritative reference to `honors-chemistry-expression-to-equation-vocabulary.md`. Solutions in `chemistry-answers.md`. Method: `chemistry-without-intuition.md`.

BEGINNER — 50 Problems (Level 1–2, 1–3 cards)

C1–C8: Stoichiometry — Mole Bridge

C1. How many moles are in $88.0\ \mathrm{g}$ of $\mathrm{CO_2}$? ($M_{\mathrm{CO_2}} = 44.0\ \mathrm{g/mol}$) [Cards: 1]

C2. What is the mass of $2.50\ \mathrm{mol}$ of $\mathrm{H_2O}$? ($M_{\mathrm{H_2O}} = 18.0\ \mathrm{g/mol}$) [Cards: 1]

C3. How many molecules are in $0.500\ \mathrm{mol}$ of $\mathrm{N_2}$? [Cards: 4, 5]

C4. What volume does $3.00\ \mathrm{mol}$ of $\mathrm{O_2}$ occupy at STP? [Cards: 6, 7]

C5. Balance and give the mole ratio of $\mathrm{H_2}$ to $\mathrm{NH_3}$: $\mathrm{N_2} + \mathrm{H_2} \rightarrow \mathrm{NH_3}$. [Cards: 8, 9]

C6. How many grams of NaCl are produced from $23.0\ \mathrm{g}$ of Na reacting with excess $\mathrm{Cl_2}$? ($M_{\mathrm{Na}} = 23.0$, $M_{\mathrm{NaCl}} = 58.5$) [Cards: 1, 8, 10]

C7. How many moles of $\mathrm{H_2O}$ are produced when $4.00\ \mathrm{mol}$ of $\mathrm{O_2}$ react with excess $\mathrm{H_2}$? ($2\mathrm{H_2} + \mathrm{O_2} \rightarrow 2\mathrm{H_2O}$) [Cards: 8, 9]

C8. Find the percent composition of carbon in $\mathrm{CH_4}$. ($M_{\mathrm{C}} = 12.0$, $M_{\mathrm{CH_4}} = 16.0$) [Cards: 17]

C9–C14: Gases

C9. Find the pressure exerted by $2.00\ \mathrm{mol}$ of an ideal gas in a $10.0\ \mathrm{L}$ container at $300\ \mathrm{K}$. ($R = 0.08206\ \mathrm{L\ atm\ mol^{-1}\ K^{-1}}$) [Cards: 19, 20]

C10. A sealed $5.00\ \mathrm{L}$ container of gas at $1.20\ \mathrm{atm}$ and $300\ \mathrm{K}$ is heated to $400\ \mathrm{K}$. What is the new pressure? [Cards: 21]

C11. A balloon at $2.00\ \mathrm{atm}$ and $300\ \mathrm{K}$ has volume $3.00\ \mathrm{L}$. It rises to where pressure is $1.50\ \mathrm{atm}$ and temperature is $280\ \mathrm{K}$. New volume? [Cards: 21]

C12. A gas at constant temperature has its pressure doubled. What happens to its volume? [Cards: 22]

C13. A gas at constant pressure is cooled from $400\ \mathrm{K}$ to $200\ \mathrm{K}$. Its initial volume is $8.00\ \mathrm{L}$. Final volume? [Cards: 23]

C14. Equal volumes of $\mathrm{H_2}$ and $\mathrm{O_2}$ at the same T and P contain equal numbers of moles. Which card guarantees this? [Cards: 25]

C15–C20: Thermochemistry Basics

C15. How much heat is required to raise $50.0\ \mathrm{g}$ of water from $20.0°\mathrm{C}$ to $80.0°\mathrm{C}$? ($c_w = 4.184\ \mathrm{J\ g^{-1}\ K^{-1}}$) [Cards: 33, 34]

C16. A reaction releases $250\ \mathrm{kJ}$ of heat to the surroundings. Is it exothermic or endothermic? What is the sign of ΔH? [Cards: 39, 40, 41]

C17. A $100.0\ \mathrm{g}$ aluminum block ($c = 0.900\ \mathrm{J\ g^{-1}\ K^{-1}}$) cools from $150.0°\mathrm{C}$ to $25.0°\mathrm{C}$. How much heat is released? [Cards: 33]

C18. In a coffee-cup calorimeter, $50.0\ \mathrm{g}$ of hot water at $60.0°\mathrm{C}$ is mixed with $50.0\ \mathrm{g}$ of cold water at $20.0°\mathrm{C}$. Predict the final temperature. ($c_w = 4.184$) [Cards: 33, 34, 36]

C19. The specific heat of water is $4.184\ \mathrm{J\ g^{-1}\ K^{-1}}$. What is the molar heat capacity of water? ($M = 18.0\ \mathrm{g/mol}$) [Cards: 34, 38]

C20. A system absorbs $500\ \mathrm{J}$ of heat and does $200\ \mathrm{J}$ of work on the surroundings. What is ΔU of the system? [Cards: 33, 39]

C21–C26: Atomic Structure & Periodicity

C21. Write the ground-state electron configuration of oxygen ($Z = 8$). [Cards: 54]

C22. What are the four quantum numbers for the last electron in sodium ($Z = 11$)? [Cards: 53, 54]

C23. How many orbitals are in the $3d$ subshell? How many electrons can it hold? [Cards: 58]

C24. Which is larger: a sodium atom (Na) or a sodium ion ($\mathrm{Na^+}$)? Why? [Cards: 59, 63]

C25. Which has the higher ionization energy: Mg or Al? Explain using the periodic trend. [Cards: 60]

C26. Arrange $\mathrm{F^-}$, Ne, and $\mathrm{Na^+}$ in order of increasing size. [Cards: 63]

C27–C32: Bonding & Molecular Structure

C27. Draw the Lewis structure of $\mathrm{NH_3}$. How many lone pairs are on the central atom? [Cards: 64]

C28. What is the formal charge on the central atom in $\mathrm{SO_4^{2-}}$ if each S–O bond is single? (Assume octets on all O atoms.) [Cards: 64, 65]

C29. Predict the molecular geometry of $\mathrm{CO_2}$. [Cards: 68, 69]

C30. What is the hybridization of the central atom in $\mathrm{BH_3}$? [Cards: 68, 70]

C31. Is $\mathrm{CO_2}$ a polar molecule? Explain. [Cards: 72, 73]

C32. Which has the higher lattice energy: NaCl or MgO? Why? [Cards: 74]

C33–C38: Kinetics Basics

C33. For the reaction $\mathrm{A} \rightarrow \textbf{products}$, doubling $[\mathrm{A}]$ doubles the rate. What is the order with respect to A? [Cards: 76, 78]

C34. A first-order reaction has $k = 0.0693\ \mathrm{s}^{-1}$. What is its half-life? [Cards: 81, 83]

C35. The half-life of a first-order reaction is $100\ \mathrm{s}$. If $[\mathrm{A}]_0 = 2.00\ \mathrm{M}$, what is $[\mathrm{A}]$ after $200\ \mathrm{s}$? [Cards: 81, 83]

C36. For a zero-order reaction, which plot is linear? [Cards: 80]

C37. The rate law is $\mathrm{Rate} = k[\mathrm{A}]^2[\mathrm{B}]$. What is the overall reaction order? [Cards: 76, 79]

C38. After three half-lives, what fraction of a radioactive sample remains? [Cards: 81, 83, 146]

C39–C44: Equilibrium Basics

C39. Write the equilibrium expression K_c for: $2\mathrm{SO_2(g)} + \mathrm{O_2(g)} \rightleftharpoons 2\mathrm{SO_3(g)}$. [Cards: 89]

C40. For $\mathrm{N_2(g)} + 3\mathrm{H_2(g)} \rightleftharpoons 2\mathrm{NH_3(g)}$, does $K_c = K_p$? Explain. [Cards: 90]

C41. If $K_c = 1.0 \times 10^3$ for a reaction, are products or reactants favored at equilibrium? [Cards: 92]

C42. A system at equilibrium has $Q = K$. If more reactant is added, what happens to Q? Which direction does the reaction shift? [Cards: 91, 97]

C43. For the exothermic reaction $2\mathrm{SO_2} + \mathrm{O_2} \rightleftharpoons 2\mathrm{SO_3}$, does increasing temperature shift equilibrium left or right? [Cards: 96, 99]

C44. For $\mathrm{N_2} + 3\mathrm{H_2} \rightleftharpoons 2\mathrm{NH_3}$, increasing pressure shifts equilibrium which way? [Cards: 96, 98]

C45–C50: Acids & Bases Basics

C45. Calculate the pH of $0.010\ \mathrm{M\ HCl}$. (HCl is a strong acid.) [Cards: 101, 107, 189]

C46. What is the pOH of a solution with $[\mathrm{OH^-}] = 1.0 \times 10^{-3}\ \mathrm{M}$? What is its pH? [Cards: 102, 103]

C47. A solution has pH = 4.50. What is $[\mathrm{H^+}]$? [Cards: 101]

C48. Write the K_a expression for acetic acid ($\mathrm{CH_3COOH}$). [Cards: 104]

C49. The K_a of a weak acid is 1.8×10^{-5}. What is its $\mathrm{p}K_a$? [Cards: 104]

C50. Calculate $[\mathrm{OH^-}]$ in a $0.20\ \mathrm{M\ NaOH}$ solution. (NaOH is a strong base.) [Cards: 108, 190]

INTERMEDIATE — 50 Problems (Level 2–3, 2–5 cards)

C51–C58: Stoichiometry — Limiting, Yield, Formulas

C51. $10.0\ \mathrm{g}$ of $\mathrm{H_2}$ reacts with $64.0\ \mathrm{g}$ of $\mathrm{O_2}$ to make water. Identify the limiting reagent and calculate the theoretical yield of water in grams. ($M_{\mathrm{H_2}} = 2.00$, $M_{\mathrm{O_2}} = 32.0$, $M_{\mathrm{H_2O}} = 18.0$) [Cards: 1, 8, 9, 11, 12]

C52. In problem C51, the actual yield of water is $25.0\ \mathrm{g}$. Calculate the percent yield. [Cards: 12, 13]

C53. $15.0\ \text{g}$ of Al reacts with $20.0\ \text{g}$ of Cl_2. ($2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$, $M_{\text{Al}} = 27.0$, $M_{\text{Cl}_2} = 71.0$, $M_{\text{AlCl}_3} = 133.5$) Find the limiting reagent and the mass of AlCl_3 produced. [Cards: 1, 8, 9, 11, 12]

C54. In problem C53, how many grams of the excess reagent remain? [Cards: 11, 12, 14]

C55. A compound is 40.0% C, 6.71% H, and 53.3% O by mass. Find its empirical formula. ($M_{\text{C}} = 12.0$, $M_{\text{H}} = 1.01$, $M_{\text{O}} = 16.0$) [Cards: 15]

C56. The empirical formula from C55 has molar mass $180\ \text{g/mol}$. What is the molecular formula? [Cards: 15, 16]

C57. Combustion of $0.500\ \text{g}$ of a hydrocarbon produces $1.65\ \text{g}$ of CO_2 and $0.675\ \text{g}$ of H_2O. Find the empirical formula. ($M_{\text{CO}_2} = 44.0$, $M_{\text{H}_2\text{O}} = 18.0$, $M_{\text{C}} = 12.0$, $M_{\text{H}} = 1.01$) [Cards: 18]

C58. A compound is found to be 92.3% C and 7.7% H by mass. Its molar mass is $78.0\ \text{g/mol}$. Determine the molecular formula. [Cards: 15, 16, 17]

C59–C66: Gases — Mixtures, Effusion, Real Gases

C59. A mixture of $2.00\ \text{mol N}_2$ and $3.00\ \text{mol O}_2$ is in a $10.0\ \text{L}$ tank at $300\ \text{K}$. Find the total pressure and the partial pressure of N_2. ($R = 0.08206$) [Cards: 19, 26, 27]

C60. Hydrogen gas is collected over water at $25°\text{C}$. The total pressure is $750\ \text{torr}$. The vapor pressure of water at $25°\text{C}$ is $23.8\ \text{torr}$. What is the partial pressure of the dry hydrogen? [Cards: 28]

C61. Calculate the ratio of effusion rates of H_2 ($M = 2.02$) to O_2 ($M = 32.0$). [Cards: 29]

C62. Calculate the root-mean-square speed of N_2 molecules at $300\ \text{K}$. ($M_{\text{N}_2} = 0.0280\ \text{kg/mol}$, $R = 8.314\ \text{J mol}^{-1}\ \text{K}^{-1}$) [Cards: 30]

C63. What is the average kinetic energy per mole of an ideal gas at $400\ \text{K}$? ($R = 8.314\ \text{J mol}^{-1}\ \text{K}^{-1}$) [Cards: 31]

C64. A gas mixture contains $0.500\ \text{mol He}$ and $1.50\ \text{mol Ne}$. The total pressure is $2.00\ \text{atm}$. Find the partial pressure of He. [Cards: 26, 27]

C65. For CO_2, $a = 3.59\ \text{L}^2\ \text{atm mol}^{-2}$ and $b = 0.0427\ \text{L mol}^{-1}$. Write the van der Waals equation for $1.00\ \text{mol}$ and explain what a and b correct for. [Cards: 32]

C66. A sealed $2.00\ \text{L}$ flask at $300\ \text{K}$ contains $0.500\ \text{mol N}_2$ and an unknown amount of Ar. The total pressure is $8.20\ \text{atm}$. How many moles of Ar are present? ($R = 0.08206$) [Cards: 19, 26]

C67–C74: Thermochemistry — Hess, Calorimetry, ΔG

C67. Calculate $\Delta H°$ for: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ using standard enthalpies of formation: $\Delta H_f^\circ(\text{CH}_4) = -74.8$, $\Delta H_f^\circ(\text{CO}_2) = -393.5$, $\Delta H_f^\circ(\text{H}_2\text{O}(\text{l})) = -285.8\ \text{kJ/mol}$. [Cards: 43]

C68. Given: $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})\ \Delta H = -393.5\ \text{kJ}$; $\text{CO(g)} + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})\ \Delta H = -283.0\ \text{kJ}$. Use Hess's Law to find ΔH for $\text{C(s)} + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO(g)}$. [Cards: 42]

C69. Estimate ΔH for $\mathrm{H_2(g) + Cl_2(g) \rightarrow 2HCl(g)}$ using bond energies: BE(H–H) = 436, BE(Cl–Cl) = 243, BE(H–Cl) = 432 kJ/mol. [Cards: 44]

C70. $0.500\ \mathrm{g}$ of methane ($\mathrm{CH_4}$) is burned in a bomb calorimeter ($C_{\mathrm{cal}} = 12.0\ \mathrm{kJ/K}$). The temperature rises by $2.08\ \mathrm{K}$. Find ΔU of combustion per mole of $\mathrm{CH_4}$. ($M_{\mathrm{CH_4}} = 16.0$) [Cards: 37]

C71. Calculate ΔG° for a reaction at $298\ \mathrm{K}$ where $\Delta H^\circ = -92.2\ \mathrm{kJ/mol}$ and $\Delta S^\circ = -198.8\ \mathrm{J\ mol^{-1}\ K^{-1}}$. Is the reaction spontaneous at $298\ \mathrm{K}$? [Cards: 48, 49]

C72. For the reaction in C71, at what temperature does $\Delta G^\circ = 0$? [Cards: 48, 52]

C73. Calculate ΔG° given $K = 1.8 \times 10^{-5}$ at $298\ \mathrm{K}$. ($R = 8.314\ \mathrm{J\ mol^{-1}\ K^{-1}}$) [Cards: 51]

C74. Calculate ΔS° for $\mathrm{N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)}$ using tabulated S° values: $S^\circ(\mathrm{N_2}) = 191.5$, $S^\circ(\mathrm{H_2}) = 130.6$, $S^\circ(\mathrm{NH_3}) = 192.3\ \mathrm{J\ mol^{-1}\ K^{-1}}$. Explain why ΔS° is negative. [Cards: 46, 47]

C75–C82: Bonding & Structure — Deeper Analysis

C75. Draw all valid resonance structures for the nitrate ion ($\mathrm{NO_3^-}$). What is the average N–O bond order? [Cards: 64, 66]

C76. Assign formal charges in the Lewis structure of CO. Which atom carries the negative formal charge? [Cards: 64, 65]

C77. Predict the molecular geometry and bond angle of $\mathrm{SF_4}$. [Cards: 68, 69, 71]

C78. What is the hybridization of the central atom in $\mathrm{PCl_5}$? [Cards: 68, 70]

C79. $\mathrm{BeCl_2}$ is linear while $\mathrm{H_2O}$ is bent. Explain using VSEPR theory. [Cards: 68, 69]

C80. Rank NaF, MgO, and KBr in order of increasing lattice energy. Explain. [Cards: 74]

C81. The bond length of $\mathrm{N \equiv N}$ is $110\ \mathrm{pm}$; $\mathrm{N = N}$ is $125\ \mathrm{pm}$; $\mathrm{N - N}$ is $145\ \mathrm{pm}$. Explain the trend. [Cards: 67, 75]

C82. $\mathrm{BF_3}$ is nonpolar despite having polar B–F bonds. Explain. [Cards: 68, 72, 73]

C83–C90: Kinetics — Integrated Rate Laws, Arrhenius, Mechanisms

C83. A reaction is first-order with $k = 0.0231\ \mathrm{min^{-1}}$. How long does it take for $[\mathrm{A}]$ to decrease from $1.00\ \mathrm{M}$ to $0.250\ \mathrm{M}$? [Cards: 81, 83]

C84. For a second-order reaction, $k = 0.540\ \mathrm{M^{-1}\ s^{-1}}$ and $[\mathrm{A}]_0 = 0.200\ \mathrm{M}$. Calculate $[\mathrm{A}]$ after $30.0\ \mathrm{s}$. [Cards: 82]

C85. A reaction has $k_1 = 0.0200\ \mathrm{s^{-1}}$ at $300\ \mathrm{K}$ and $k_2 = 0.0800\ \mathrm{s^{-1}}$ at $320\ \mathrm{K}$. Calculate E_a. ($R = 8.314\ \mathrm{J\ mol^{-1}\ K^{-1}}$) [Cards: 85]

C86. The activation energy of a reaction is $75.0\ \mathrm{kJ/mol}$. By what factor does the rate increase when the temperature is raised from $300\ \mathrm{K}$ to $310\ \mathrm{K}$? ($R = 8.314$) [Cards: 84, 85]

C87. A proposed mechanism: Step 1 (slow): $\mathrm{A + B \rightarrow C}$. Step 2 (fast): $\mathrm{C + A \rightarrow D}$. What is the overall reaction? What is the rate law? [Cards: 76, 87]

C88. How does a catalyst increase the rate of a reaction? Does it change ΔH, ΔG, or K? [Cards: 86, 88]

C89. The following kinetic data were obtained for $\mathrm{A} + \mathrm{B} \rightarrow \text{products}$:

Trial	$[\mathrm{A}]_0$ (M)	$[\mathrm{B}]_0$ (M)	Initial Rate (M/s)
1	0.100	0.100	4.0×10^{-5}
2	0.200	0.100	1.6×10^{-4}
3	0.100	0.200	8.0×10^{-5}

Determine the rate law and the value of k with units. [Cards: 76, 77, 78]

C90. A zero-order reaction has $k = 0.0150\ \mathrm{M\ s^{-1}}$ and $[\mathrm{A}]_0 = 0.600\ \mathrm{M}$. How long until the reactant is completely consumed? [Cards: 80]

C91–C98: Equilibrium — ICE Tables & Applications

C91. For $\mathrm{H_2(g)} + \mathrm{I_2(g)} \rightleftharpoons \mathrm{2HI(g)}$, $K_c = 50.0$ at $700\ \mathrm{K}$. Initially $[\mathrm{H_2}] = [\mathrm{I_2}] = 1.00\ \mathrm{M}$, $[\mathrm{HI}] = 0$. Set up the ICE table and write the expression to solve for equilibrium concentrations. [Cards: 89, 93, 95]

C92. For a reaction with $K_c = 1.0 \times 10^{-4}$, the initial concentrations are $[\mathrm{A}]_0 = 0.500\ \mathrm{M}$, $[\mathrm{B}]_0 = 0$. Can the small-K approximation be used? If so, find $[\mathrm{B}]$ at equilibrium for $\mathrm{A} \rightleftharpoons \mathrm{2B}$. [Cards: 89, 93, 94]

C93. For $\mathrm{PCl_5(g)} \rightleftharpoons \mathrm{PCl_3(g)} + \mathrm{Cl_2(g)}$, $K_p = 1.05$ at $250°\mathrm{C}$. What is K_c at this temperature? ($R = 0.08206$) [Cards: 90]

C94. At a certain moment, $[\mathrm{SO_2}] = 0.20\ \mathrm{M}$, $[\mathrm{O_2}] = 0.10\ \mathrm{M}$, $[\mathrm{SO_3}] = 0.40\ \mathrm{M}$ for $\mathrm{2SO_2} + \mathrm{O_2} \rightleftharpoons \mathrm{2SO_3}$ with $K_c = 4.3 \times 10^2$. Calculate Q and predict the direction the reaction must shift to reach equilibrium. [Cards: 89, 91]

C95. For the exothermic reaction $\mathrm{N_2} + \mathrm{3H_2} \rightleftharpoons \mathrm{2NH_3}$, predict the effect of (a) adding $\mathrm{N_2}$, (b) decreasing volume, (c) increasing temperature on the equilibrium position. [Cards: 96, 97, 98, 99]

C96. $0.500\ \mathrm{mol}$ of HI is placed in a $1.00\ \mathrm{L}$ flask at $520\ \mathrm{K}$. At equilibrium, $[\mathrm{HI}] = 0.400\ \mathrm{M}$. Calculate K_c for $\mathrm{2HI(g)} \rightleftharpoons \mathrm{H_2(g)} + \mathrm{I_2(g)}$. [Cards: 89, 93, 95]

C97. For $\mathrm{N_2O_4(g)} \rightleftharpoons \mathrm{2NO_2(g)}$, $K_p = 0.148$ at $298\ \mathrm{K}$. Initially $P_{\mathrm{N_2O_4}} = 1.00\ \mathrm{atm}$, $P_{\mathrm{NO_2}} = 0$. Find the equilibrium partial pressures. (Solve the quadratic — do NOT use approximation.) [Cards: 89, 93, 95]

C98. $K_c = 0.212$ at $100°\mathrm{C}$ for $\mathrm{2NOBr(g)} \rightleftharpoons \mathrm{2NO(g)} + \mathrm{Br_2(g)}$. In a $2.00\ \mathrm{L}$ flask, $0.400\ \mathrm{mol\ NOBr}$, $0.200\ \mathrm{mol\ NO}$, and $0.300\ \mathrm{mol\ Br_2}$ are mixed. Is the system at equilibrium? If not, which way does it shift? [Cards: 89, 91, 93]

C99–C100: Cross-Domain Bridges

C99. $5.00\ \mathrm{g}$ of $\mathrm{CaCO_3}$ ($M = 100.1$) is heated to decompose: $\mathrm{CaCO_3(s)} \rightarrow \mathrm{CaO(s)} + \mathrm{CO_2(g)}$. What volume of $\mathrm{CO_2}$ is produced at STP? [Cards: 1, 6, 7, 8, 9, 160]

C100. The combustion of $2.00\ \mathrm{g}$ of propane ($\mathrm{C_3H_8}$, $M = 44.1$) releases $101\ \mathrm{kJ}$ of heat ($\Delta H_{\mathrm{comb}} = -2220\ \mathrm{kJ/mol}$). This heat is absorbed by $500\ \mathrm{g}$ of water initially at $22.0°\mathrm{C}$. Find the final temperature of the water. ($c_w = 4.184$) [Cards: 1, 33, 34, 162]

ADVANCED — 50 Problems (Level 3–4, 4–10 cards)

C101–C110: Acids & Bases — Buffers, Titrations, Polyprotic

C101. Calculate the pH of $0.100\ \mathrm{M\ CH_3COOH}$ ($K_a = 1.8 \times 10^{-5}$). [Cards: 101, 104, 109]

C102. Calculate the pH of $0.100\ \mathrm{M\ NH_3}$ ($K_b = 1.8 \times 10^{-5}$). [Cards: 101, 102, 105, 110]

C103. A buffer is prepared by mixing $0.500\ \mathrm{mol}$ of acetic acid ($\mathrm{CH_3COOH}$) and $0.500\ \mathrm{mol}$ of sodium acetate ($\mathrm{CH_3COONa}$) in $1.00\ \mathrm{L}$ of solution. $K_a = 1.8 \times 10^{-5}$. Calculate the pH. [Cards: 104, 111, 112, 113, 169]

C104. To the buffer in C103, $0.100\ \mathrm{mol}$ of HCl is added (assume no volume change). Calculate the new pH. [Cards: 104, 111, 112, 169]

C105. A $25.0\ \mathrm{mL}$ sample of $0.100\ \mathrm{M\ HCl}$ is titrated with $0.100\ \mathrm{M\ NaOH}$. Calculate the pH after adding (a) $0\ \mathrm{mL}$, (b) $12.5\ \mathrm{mL}$, (c) $25.0\ \mathrm{mL}$, (d) $30.0\ \mathrm{mL}$ of NaOH. [Cards: 101, 107, 114]

C106. A $50.0\ \mathrm{mL}$ sample of $0.100\ \mathrm{M\ CH_3COOH}$ ($K_a = 1.8 \times 10^{-5}$) is titrated with $0.100\ \mathrm{M\ NaOH}$. Calculate the pH at (a) $0\ \mathrm{mL}$, (b) $25.0\ \mathrm{mL}$ (half-equivalence), (c) $50.0\ \mathrm{mL}$ (equivalence) of NaOH added. [Cards: 101, 104, 109, 112, 114, 115, 116, 170]

C107. $0.200\ \mathrm{M}$ sodium acetate ($\mathrm{CH_3COONa}$) dissolves in water. Calculate the pH of the solution. (K_a of $\mathrm{CH_3COOH} = 1.8 \times 10^{-5}$) [Cards: 101, 104, 105, 119]

C108. Calculate the pH of $0.100\ \mathrm{M\ H_2CO_3}$ (a diprotic acid with $K_{a1} = 4.3 \times 10^{-7}$, $K_{a2} = 5.6 \times 10^{-11}$). [Cards: 101, 104, 109, 118]

C109. What is the percent ionization of $0.100\ \mathrm{M\ CH_3COOH}$ ($K_a = 1.8 \times 10^{-5}$)? What is the percent ionization if the solution is diluted to $0.0100\ \mathrm{M}$? [Cards: 101, 104, 106, 109]

C110. A buffer is needed at pH 9.25. Which acid/conjugate-base pair is best: (a) $\mathrm{CH_3COOH/CH_3COO^-}$ ($\mathrm{p}K_a = 4.74$), (b) $\mathrm{NH_4^+/NH_3}$ ($\mathrm{p}K_a = 9.25$), (c) $\mathrm{H_2PO_4^-/HPO_4^{2-}}$ ($\mathrm{p}K_a = 7.21$)? What ratio of $[\mathrm{A^-}]/[\mathrm{HA}]$ is needed? [Cards: 104, 112, 113, 169]

C111–C120: Electrochemistry — Cells, Nernst, Electrolysis

C111. For the reaction $\mathrm{Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)}$, given $E^\circ_{\mathrm{Zn^{2+}/Zn}} = -0.76\ \mathrm{V}$ and $E^\circ_{\mathrm{Cu^{2+}/Cu}} = +0.34\ \mathrm{V}$, calculate E°_{cell}. Is the reaction spontaneous? [Cards: 121, 122, 125, 126]

C112. For the cell in C111, calculate ΔG° and K at $298\ \mathrm{K}$. ($F = 96{,}485\ \mathrm{C/mol\ e^-}$, $R = 8.314$) [Cards: 51, 125, 127, 129]

C113. Calculate the cell potential at $298\ \mathrm{K}$ for: $\mathrm{Zn(s)|Zn^{2+}(0.010\ M)||Cu^{2+}(1.00\ M)|Cu(s)}$. $E^\circ_{\mathrm{cell}} = 1.10\ \mathrm{V}$. [Cards: 125, 128]

C114. A current of $2.50\ \mathrm{A}$ is passed through molten NaCl for $30.0\ \mathrm{minutes}$. What mass of sodium metal is produced? ($M_{\mathrm{Na}} = 23.0$, $F = 96{,}485$) [Cards: 129, 131, 132]

C115. Balance the following redox reaction in acidic solution: $\mathrm{MnO_4^-} + \mathrm{Fe^{2+}} \rightarrow \mathrm{Mn^{2+}} + \mathrm{Fe^{3+}}$. [Cards: 121, 122, 123, 124]

C116. For the reaction $2\mathrm{Al(s)} + 3\mathrm{I_2(s)} \rightarrow 2\mathrm{Al^{3+}(aq)} + 6\mathrm{I^-(aq)}$, $E^\circ_{\mathrm{cell}} = 2.20\ \mathrm{V}$. Calculate ΔG° and determine K at $298\ \mathrm{K}$. [Cards: 51, 125, 127, 129, 166]

C117. An electrolytic cell deposits $1.50\ \mathrm{g}$ of Cu from $\mathrm{CuSO_4}$ solution using a current of $1.80\ \mathrm{A}$. How long did the electrolysis take? ($M_{\mathrm{Cu}} = 63.5$, $F = 96{,}485$) [Cards: 129, 131, 132]

C118. Calculate E°_{cell} for: $2\mathrm{Fe^{3+}(aq)} + 2\mathrm{I^-(aq)} \rightarrow 2\mathrm{Fe^{2+}(aq)} + \mathrm{I_2(s)}$. Given: $E^\circ_{\mathrm{Fe^{3+}/Fe^{2+}}} = +0.77\ \mathrm{V}$, $E^\circ_{\mathrm{I_2/I^-}} = +0.54\ \mathrm{V}$. Is this reaction spontaneous under standard conditions? [Cards: 121, 122, 125, 126]

C119. For the cell $\mathrm{Pt}|\mathrm{H_2}(1\ \mathrm{atm})|\mathrm{H^+}(\mathrm{pH} =?)||\mathrm{Cu^{2+}}(1.00\ \mathrm{M})|\mathrm{Cu(s)}$, the measured potential is $0.480\ \mathrm{V}$. Given $E^\circ_{\mathrm{cell}} = 0.34\ \mathrm{V}$, find the pH of the anode compartment. [Cards: 101, 125, 128]

C120. A galvanic cell uses the reaction $\mathrm{Cd(s)} + \mathrm{Ni^{2+}(aq)} \rightarrow \mathrm{Cd^{2+}(aq)} + \mathrm{Ni(s)}$ with $E^\circ_{\mathrm{cell}} = 0.15\ \mathrm{V}$. Initially $[\mathrm{Ni^{2+}}] = 1.00\ \mathrm{M}$, $[\mathrm{Cd^{2+}}] = 0.0100\ \mathrm{M}$. Calculate the initial cell potential and predict whether it will increase or decrease as the cell discharges. [Cards: 125, 126, 128]

C121–C130: Solutions — Colligative Properties & Concentration

C121. Calculate the molality of a solution made by dissolving $10.0\ \mathrm{g}$ of NaCl ($M = 58.5$) in $200\ \mathrm{g}$ of water. [Cards: 134]

C122. What is the boiling point of an aqueous solution containing $34.2\ \mathrm{g}$ of sucrose ($\mathrm{C_{12}H_{22}O_{11}}$, $M = 342$) in $500\ \mathrm{g}$ of water? ($K_b = 0.512\ \mathrm{K\ kg\ mol^{-1}}$, $i = 1$ for sucrose) [Cards: 134, 141, 144]

C123. Calculate the freezing point of a solution of $5.85\ \mathrm{g}$ of NaCl ($M = 58.5$) dissolved in $100\ \mathrm{g}$ of water. ($K_f = 1.86\ \mathrm{K\ kg\ mol^{-1}}$) [Cards: 134, 142, 144]

C124. What is the osmotic pressure at $298\ \mathrm{K}$ of a $0.0100\ \mathrm{M}$ aqueous solution of $\mathrm{CaCl_2}$ (assume complete dissociation)? ($R = 0.08206\ \mathrm{L\ atm\ mol^{-1}\ K^{-1}}$) [Cards: 143, 144]

C125. The vapor pressure of pure water at $25^\circ\mathrm{C}$ is $23.8\ \mathrm{torr}$. What is the vapor pressure of a solution containing $18.0\ \mathrm{g}$ of glucose ($\mathrm{C_6H_{12}O_6}$, $M = 180$) in $90.0\ \mathrm{g}$ of water? ($M_{\mathrm{H_2O}} = 18.0$) [Cards: 136, 140]

C126. What concentration of NaCl (in mol/L) is isotonic with blood, which has an osmotic pressure of $7.7\ \mathrm{atm}$ at $310\ \mathrm{K}$? ($R = 0.08206$, assume $i = 2$) [Cards: 143, 144]

C127. $25.0\ \mathrm{mL}$ of $6.00\ \mathrm{M\ HCl}$ is diluted to $500\ \mathrm{mL}$. What is the new concentration? [Cards: 133, 137]

C128. A solution is prepared by mixing $50.0\ \mathrm{g}$ of ethanol ($\mathrm{C_2H_5OH}$, $M = 46.1$) with $50.0\ \mathrm{g}$ of water ($M = 18.0$). Calculate the mole fraction of each component. [Cards: 136]

C129. Rank the following aqueous solutions in order of increasing freezing point: $0.10\ \mathrm{m\ NaCl}$, $0.10\ \mathrm{m\ CaCl_2}$, $0.10\ \mathrm{m\ C_6H_{12}O_6}$, $0.10\ \mathrm{m\ CH_3COOH}$ (weak acid, assume minimal ionization). [Cards: 139, 142, 144]

C130. A $0.500\ \mathrm{g}$ protein is dissolved in enough water to make $50.0\ \mathrm{mL}$ of solution. The osmotic pressure is $3.85\ \mathrm{torr}$ at $298\ \mathrm{K}$. What is the molar mass of the protein? ($R = 0.08206\ \mathrm{L\ atm\ mol^{-1}\ K^{-1}}$, $1\ \mathrm{atm} = 760\ \mathrm{torr}$) [Cards: 133, 143, 144, 199]

C131–C140: Nuclear Chemistry

C131. The half-life of $^{14}\mathrm{C}$ is $5730\ \mathrm{years}$. A sample contains 25% of the original $^{14}\mathrm{C}$. How old is the sample? [Cards: 145, 146]

C132. $^{238}\mathrm{U}$ decays by alpha emission. Write the balanced nuclear equation and identify the daughter nuclide. [Cards: 148]

C133. $^{131}\mathrm{I}$ ($t_{1/2} = 8.02\ \mathrm{days}$) is used in medical imaging. What fraction remains after $30.0\ \mathrm{days}$? [Cards: 145, 146]

C134. Calculate the binding energy per nucleon of $^{56}\mathrm{Fe}$ (mass = $55.9349\ \mathrm{u}$). ($m_p = 1.00728\ \mathrm{u}$, $m_n = 1.00867\ \mathrm{u}$, $1\ \mathrm{u} = 931.5\ \mathrm{MeV}/c^2$) [Cards: 153, 154, 155, 207]

C135. $^{60}\mathrm{Co}$ decays by β^- emission with $t_{1/2} = 5.27\ \mathrm{years}$. A sample has an activity of $2.50 \times 10^4\ \mathrm{Bq}$. What was the activity $10.0\ \mathrm{years}$ ago? [Cards: 145, 146, 147, 149]

C136. Complete the nuclear equation: $^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + ?$. What type of decay is this? [Cards: 148]

C137. A rock contains $0.500\ \mathrm{g}$ of $^{238}\mathrm{U}$ ($t_{1/2} = 4.47 \times 10^9\ \mathrm{yr}$) and $0.125\ \mathrm{g}$ of $^{206}\mathrm{Pb}$. Assuming all the lead came from uranium decay, estimate the age of the rock. ($M_{238} = 238$, $M_{206} = 206$) [Cards: 1, 145, 146]

C138. Calculate the mass defect and binding energy of $^{4}\mathrm{He}$ (mass = $4.00260\ \mathrm{u}$). ($m_p = 1.00728\ \mathrm{u}$, $m_n = 1.00867\ \mathrm{u}$, $1\ \mathrm{u} = 931.5\ \mathrm{MeV}/c^2$) [Cards: 153, 154, 207]

C139. The decay constant of $^{32}\mathrm{P}$ is $0.0485\ \mathrm{day}^{-1}$. Calculate its half-life and the time for 90% to decay. [Cards: 145, 146]

C140. $^{235}\mathrm{U}$ ($t_{1/2} = 7.04 \times 10^8\ \mathrm{yr}$) and $^{238}\mathrm{U}$ ($t_{1/2} = 4.47 \times 10^9\ \mathrm{yr}$) are both present in natural uranium. Today, $^{235}\mathrm{U}$ is 0.72% of natural uranium. If the isotopes were originally equally abundant, how long ago did they form? (Solve using the ratio of decay equations.) [Cards: 145, 146]

C141–C150: Multi-Domain Synthesis — Olympiad Level

C141. $10.0\ \mathrm{g}$ of impure $\mathrm{CaCO_3}$ ($M = 100.1$) is treated with excess HCl. The $\mathrm{CO_2}$ produced is collected over water at $25^\circ\mathrm{C}$ ($P_{\mathrm{H_2O}} = 23.8\ \mathrm{torr}$). The volume of gas collected is $2.15\ \mathrm{L}$ at a total pressure of $755\ \mathrm{torr}$. What is the percent purity of the $\mathrm{CaCO_3}$? ($R = 0.08206$) [Cards: 1, 8, 9, 19, 20, 28, 160]

C142. $0.500\ \mathrm{mol}$ of $\mathrm{N_2O_4}$ is placed in a $5.00\ \mathrm{L}$ flask at $298\ \mathrm{K}$. At equilibrium, the total pressure is $3.20\ \mathrm{atm}$. Calculate K_p and K_c for $\mathrm{N_2O_4(g)} \rightleftharpoons \mathrm{2NO_2(g)}$. ($R = 0.08206$) [Cards: 19, 89, 90, 93, 95]

C143. A $0.250\ \mathrm{M}$ solution of a weak monoprotic acid has pH $= 4.25$. Calculate K_a and the percent ionization. What is the freezing point depression of this solution? ($K_f = 1.86\ \mathrm{K\ kg\ mol^{-1}}$, assume density $\approx 1.00\ \mathrm{g/mL}$, neglect ion pairing.) [Cards: 101, 104, 106, 109, 134, 142, 144]

C144. The reaction $\mathrm{A} \rightarrow \mathrm{products}$ is first-order. At $300\ \mathrm{K}$, $k = 0.0200\ \mathrm{min}^{-1}$. At $320\ \mathrm{K}$, $k = 0.0600\ \mathrm{min}^{-1}$. Calculate E_a and predict k at $340\ \mathrm{K}$. If $[\mathrm{A}]_0 = 1.50\ \mathrm{M}$, how long at $340\ \mathrm{K}$ to reach

$0.150\ \mathrm{M}$? [Cards: 81, 83, 84, 85, 86]

C145. A voltaic cell is constructed: $\mathrm{Mg(s)|Mg^{2+}(0.050\ M)||Ag^{+}(0.010\ M)|Ag(s)}$. $E^{\circ}_{\mathrm{Mg^{2+}/Mg}} = -2.37\ \mathrm{V}$, $E^{\circ}_{\mathrm{Ag^{+}/Ag}} = +0.80\ \mathrm{V}$. Calculate $E^{\circ}_{\mathrm{cell}}$, E_{cell} at the given concentrations, ΔG, and K at $298\ \mathrm{K}$. ($F = 96{,}485$, $R = 8.314$) [Cards: 51, 125, 126, 127, 128, 165, 166]

C146. $2.00\ \mathrm{g}$ of benzoic acid ($\mathrm{C_6H_5COOH}$, $M = 122$) is dissolved in $50.0\ \mathrm{g}$ of benzene. The freezing point is lowered by $1.62\ \mathrm{K}$. K_f for benzene is $5.12\ \mathrm{K\ kg\ mol^{-1}}$. Benzoic acid dimerizes in benzene: $\mathrm{2C_6H_5COOH \rightleftharpoons (C_6H_5COOH)_2}$. Calculate the equilibrium constant K_c for the dimerization. (Assume 1 L of benzene solution ≈ 1 kg for concentration calculation.) [Cards: 89, 93, 95, 134, 142, 144]

C147. For the reaction $\mathrm{2NO(g) + O_2(g) \rightarrow 2NO_2(g)}$, the rate law is $\mathrm{Rate} = k[\mathrm{NO}]^2[\mathrm{O_2}]$. At $298\ \mathrm{K}$, $k = 7.1 \times 10^3\ \mathrm{M^{-2}\ s^{-1}}$. An industrial reactor initially has $P_{\mathrm{NO}} = 0.500\ \mathrm{atm}$ and $P_{\mathrm{O_2}} = 0.250\ \mathrm{atm}$ at $400\ \mathrm{K}$ in a $10.0\ \mathrm{L}$ vessel. Find the initial rate in M/s. ($R = 0.08206$) [Cards: 19, 76, 77, 133]

C148. $1.00\ \mathrm{L}$ of a buffer contains $0.200\ \mathrm{mol\ CH_3COOH}$ and $0.300\ \mathrm{mol\ CH_3COONa}$. ($K_a = 1.8 \times 10^{-5}$) (a) Calculate the initial pH. (b) $0.0500\ \mathrm{mol}$ of NaOH is added. Calculate the new pH. (c) $0.0500\ \mathrm{mol}$ of HCl is added (to the original buffer). Calculate the new pH. (d) Compare the pH changes — is the buffer effective? [Cards: 101, 104, 111, 112, 113, 169]

C149. The solubility product of AgCl is $K_{sp} = 1.8 \times 10^{-10}$. $\mathrm{Ag^+}$ forms a complex with $\mathrm{NH_3}$: $\mathrm{Ag^+ + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+}$ with $K_f = 1.7 \times 10^7$. Calculate the molar solubility of AgCl in $1.0\ \mathrm{M\ NH_3}$. (Hint: Combine the two equilibria; the very large K_f means you may assume nearly complete complexation.) [Cards: 89, 93, 95]

C150. Consider the reaction $\mathrm{A(g) \rightleftharpoons B(g) + C(g)}$ in a $2.00\ \mathrm{L}$ flask at $500\ \mathrm{K}$. Initially $P_{\mathrm{A}} = 1.50\ \mathrm{atm}$. At equilibrium, the total pressure is $2.10\ \mathrm{atm}$. (a) Calculate K_p. (b) ΔG° for this reaction at $500\ \mathrm{K}$. (c) If $\Delta H^{\circ} = 42.0\ \mathrm{kJ/mol}$ (constant over temperature), calculate K_p at $600\ \mathrm{K}$. (d) Predict the equilibrium shift if the volume is halved. ($R = 8.314\ \mathrm{J\ mol^{-1}\ K^{-1}}$) [Cards: 19, 48, 51, 89, 90, 93, 95, 96, 98, 99, 164]

INDUSTRIAL & APPLIED CHEMISTRY — 30 Problems (Level 2–4, 2–8 cards)

> Real processes. Real numbers. Same cards. The Haber plant, the chlor-alkali cell, the oil refinery — they all obey the same equations you already know. The only difference is that the "given" values sound like an engineer's clipboard instead of a textbook.

C151–C158: Haber-Bosch Process — Ammonia Synthesis

C151. The Haber process: $\mathrm{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)}$, $\Delta H^{\circ} = -92.2\ \mathrm{kJ/mol}$. A reactor at $450^{\circ}\mathrm{C}$ and $200\ \mathrm{atm}$ uses a Fe/Ru catalyst. (a) Explain why high pressure favors ammonia. (b) Explain why the reaction is run at $450^{\circ}\mathrm{C}$ despite being exothermic. (c) What does the catalyst do — and what does it NOT do? [Cards: 40, 86, 88, 96, 98, 99]

C152. A Haber reactor is fed $100\ \mathrm{mol/min}$ of $\mathrm{N_2}$ and $300\ \mathrm{mol/min}$ of $\mathrm{H_2}$. At equilibrium at $400^{\circ}\mathrm{C}$, the mole fraction of $\mathrm{NH_3}$ in the exit stream is 0.25. What is the single-pass conversion of $\mathrm{N_2}$? [Cards: 8, 9, 93, 95]

C153. In the Haber process, $\Delta G^\circ = -33.0\ \mathrm{kJ/mol}$ at $298\ \mathrm{K}$ (see C71). At the operating temperature of $723\ \mathrm{K}$, $\Delta G^\circ \approx +20\ \mathrm{kJ/mol}$ (nonspontaneous under standard conditions). Explain why the industrial process still works. Calculate K_p at $723\ \mathrm{K}$ from ΔG°. ($R = 8.314$) [Cards: 48, 49, 51, 99]

C154. A Haber plant produces $1000\ \text{metric tons}$ of $\mathrm{NH_3}$ per day ($1\ \text{ton} = 10^6\ \mathrm{g}$). What mass of $\mathrm{N_2}$ is consumed daily, assuming 95% overall yield (including recycle)? ($M_{\mathrm{NH_3}} = 17.0$, $M_{\mathrm{N_2}} = 28.0$) [Cards: 1, 8, 9, 10, 13]

C155. The activation energy of ammonia synthesis on an iron catalyst is $160\ \mathrm{kJ/mol}$ (uncatalyzed: $940\ \mathrm{kJ/mol}$). By what factor does the catalyst increase the rate at $723\ \mathrm{K}$? ($R = 8.314$) [Cards: 84, 85, 86, 88]

C156. ΔS° for the Haber reaction is $-198.8\ \mathrm{J\ mol^{-1}\ K^{-1}}$ (see C74). The plant engineer notes that at $723\ \mathrm{K}$, the ΔG° calculated from $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ is $+20\ \mathrm{kJ/mol}$. Verify this calculation given $\Delta H^\circ = -92.2\ \mathrm{kJ/mol}$. Why does the entropy term dominate at high T? [Cards: 46, 48, 52]

C157. The feed to a Haber reactor is a $3:1\ \mathrm{H_2:N_2}$ mixture at $500\ \mathrm{K}$ and $300\ \mathrm{atm}$ total pressure. At equilibrium, the partial pressure of $\mathrm{NH_3}$ is $60.0\ \mathrm{atm}$. Calculate K_p. (Hint: use Dalton's Law to find $P_{\mathrm{N_2}}$ and $P_{\mathrm{H_2}}$.) [Cards: 19, 26, 27, 89, 93, 95]

C158. The Haber process uses a recycle loop: unreacted $\mathrm{N_2}$ and $\mathrm{H_2}$ are separated from the ammonia product and returned to the reactor. If the single-pass conversion of $\mathrm{N_2}$ is 15%, how many passes through the reactor are needed (theoretically) for 99% overall conversion? (Treat each pass as independent; use $(1 - 0.15)^n = 0.01$.) [Cards: 81, 145]

C159–C164: Contact Process — Sulfuric Acid

C159. The Contact process step: $\mathrm{2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)}$, $\Delta H^\circ = -197.8\ \mathrm{kJ/mol}$. At $450^\circ\mathrm{C}$, $K_p = 4.3 \times 10^2$. A reactor initially has $P_{\mathrm{SO_2}} = 1.00\ \mathrm{atm}$, $P_{\mathrm{O_2}} = 0.50\ \mathrm{atm}$, $P_{\mathrm{SO_3}} = 0$. Calculate the equilibrium partial pressures of all species. [Cards: 89, 93, 95]

C160. In the Contact process, the $\mathrm{SO_2}$ oxidation is run at $450^\circ\mathrm{C}$ with a $\mathrm{V_2O_5}$ catalyst. At $298\ \mathrm{K}$, $K_p \approx 10^{25}$, but the reaction is kinetically frozen (rates are negligible). Explain the trade-off between thermodynamics (high K at low T) and kinetics (reasonable rate only at high T). [Cards: 40, 84, 86, 88, 92, 99]

C161. Roasting of pyrite ($\mathrm{FeS_2}$) produces $\mathrm{SO_2}$ for the Contact process: $\mathrm{4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2}$. A roaster processes $500\ \mathrm{kg}$ of pyrite (assume $100\%\ \mathrm{FeS_2}$, $M = 120$) per hour. What volume of $\mathrm{SO_2}$ at STP is produced per hour? [Cards: 1, 6, 7, 8, 9, 160]

C162. The $\mathrm{SO_3}$ from the Contact process is absorbed in $98\%\ \mathrm{H_2SO_4}$ (not water — to avoid acid mist). The absorption reaction: $\mathrm{SO_3 + H_2SO_4 \rightarrow H_2S_2O_7}$ (oleum), then $\mathrm{H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4}$. If a plant absorbs $1000\ \mathrm{kg}$ of $\mathrm{SO_3}$ ($M = 80.1$) per hour, what mass of $100\%\ \mathrm{H_2SO_4}$ ($M = 98.1$) is ultimately produced? (The net is $\mathrm{SO_3 + H_2O \rightarrow H_2SO_4}$.) [Cards: 1, 8, 9, 10]

C163. $\mathrm{V_2O_5}$ catalyzes $\mathrm{SO_2}$ oxidation via a two-step redox mechanism: $\mathrm{V_2O_5 + SO_2 \rightarrow V_2O_4 + SO_3}$, then $\mathrm{V_2O_4 + \frac{1}{2}O_2 \rightarrow V_2O_5}$. (a) What is the net reaction? (b) Why is $\mathrm{V_2O_5}$ a catalyst rather than a reactant? (c) Assign oxidation numbers to V in $\mathrm{V_2O_5}$ and $\mathrm{V_2O_4}$. [Cards: 88, 121, 122, 123]

C164. An Contact process plant operates at 98% conversion of SO_2 to SO_3, and 99.5% absorption of SO_3 into H_2SO_4. If the plant feeds 200 kg of SO_2 ($M = 64.1$) per hour, what is the daily production of H_2SO_4 ($M = 98.1$) in metric tons? Assume 24-hour operation. [Cards: 1, 8, 9, 10, 13]

C165–C170: Chlor-Alkali & Electrochemical Industry

C165. The chlor-alkali process electrolyzes brine ($NaCl$ solution): $2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$. A membrane cell operates at 4.0 V with $150{,}000$ A current. What mass of $NaOH$ ($M = 40.0$) is produced per hour? ($F = 96{,}485$) [Cards: 121, 122, 123, 129, 131, 132]

C166. The chlor-alkali cell in C165 has a current efficiency of 96% (some current is lost to side reactions). What is the actual daily production of Cl_2 ($M = 71.0$) in metric tons? The half-reaction is $2Cl^- \rightarrow Cl_2 + 2e^-$. [Cards: 129, 131, 132]

C167. Aluminum is produced by the Hall-Héroult process: Al_2O_3 (dissolved in cryolite) is electrolyzed. The half-reaction at the cathode: $Al^{3+} + 3e^- \rightarrow Al$. A smelter operates at $250{,}000$ A. How many kg of Al ($M = 27.0$) are produced per cell per day? ($F = 96{,}485$) [Cards: 129, 131, 132]

C168. In the Hall-Héroult process, the anode reaction consumes the carbon anode: $C(s) + 2O^{2-} \rightarrow CO_2(g) + 4e^-$. For every 1.00 kg of Al produced, what mass of carbon ($M = 12.0$) is consumed? ($M_{Al} = 27.0$) [Cards: 1, 8, 9, 10, 131]

C169. A lithium-ion battery cell has a nominal voltage of 3.7 V and capacity of 3000 mAh (milliamp-hours). (a) Convert the capacity to coulombs. (b) If the cell reaction transfers 1 mole of electrons per mole of Li^+, how many grams of lithium ($M = 6.94$) are theoretically moved between electrodes during a full discharge? ($F = 96{,}485$) [Cards: 129, 131, 132]

C170. Electroplating: a factory plates chromium onto steel bumpers from a CrO_3 bath (Cr is reduced from Cr^{6+} to Cr^0). A bumper requires 2.00 g of Cr ($M = 52.0$) deposited. The plating bath operates at 500 A. How many seconds per bumper? ($F = 96{,}485$) [Cards: 121, 122, 129, 131, 132]

C171–C176: Industrial Thermochemistry & Fuels

C171. A coal-fired power plant burns C (coal approximated as pure carbon) at a rate of 2500 metric tons/day. $C(s) + O_2(g) \rightarrow CO_2(g)$, $\Delta H^\circ = -393.5$ kJ/mol. What is the daily heat output in GJ? ($M_C = 12.0$, $1\text{ GJ} = 10^9$ J) [Cards: 1, 39, 40]

C172. The CO_2 from C171 is captured using an amine scrubber that absorbs 90% of the CO_2. What mass of CO_2 ($M = 44.0$) is captured per day? What volume would this CO_2 occupy at $25^\circ C$ and 1.00 atm? ($R = 0.08206$) [Cards: 1, 8, 9, 10, 19]

C173. Lime is produced by thermal decomposition of limestone: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$, $\Delta H^\circ = +178$ kJ/mol. A lime kiln fires natural gas (CH_4, $\Delta H_{comb} = -890$ kJ/mol) to supply the heat. How many kg of CH_4 ($M = 16.0$) are needed to decompose 1000 kg of $CaCO_3$ ($M = 100.1$), assuming 60% thermal efficiency? [Cards: 1, 8, 9, 10, 39, 40, 41, 162]

C174. Syngas ($CO + H_2$) is produced by steam reforming: $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$, $\Delta H^\circ = +206$ kJ/mol. The reaction is run at $800^\circ C$ and 25 atm. (a) Why high temperature? (b) Why is excess steam (a reactant) used despite the high T already favoring products? (c) If the reactor produces

$1000\ m^3$ of syngas (measured at STP) per hour, how many moles of H_2 are in this stream if the $CO : H_2$ ratio is $1 : 3$? [Cards: 6, 7, 19, 26, 39, 41, 96, 97, 99]

C175. Water-gas shift reaction: $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$, $\Delta H^\circ = -41.2\ kJ/mol$. At $500\ K$, $K_p = 126$. In an industrial shift reactor at $500\ K$, the inlet gas contains $1.00\ mol\ CO$, $3.00\ mol\ H_2O$, $0.50\ mol\ CO_2$, and $0.50\ mol\ H_2$ (per unit volume). (a) Calculate Q. (b) Which direction does the reaction proceed? (c) At equilibrium, 99% of CO has reacted. Verify this with an ICE table. [Cards: 89, 91, 93, 95]

C176. Methanol synthesis: $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$, $\Delta H^\circ = -90.7\ kJ/mol$. At $500\ K$, $K_p = 6.1 \times 10^{-3}$. Why is this reaction run at high pressure (50–$100\ atm$) despite the catalyst (Cu/ZnO) working best at only $250^\circ C$? Use Le Chatelier's principle and the Arrhenius concept. [Cards: 40, 84, 86, 88, 96, 98, 99]

C177–C180: Polymer, Environmental & Materials Industry

C177. Polyethylene is produced by polymerization of ethylene: $nC_2H_4 \rightarrow (C_2H_4)_n$. If a reactor polymerizes $1000\ kg$ of ethylene ($M = 28.1$) at 98% conversion, what mass of polyethylene is produced? (Polymerization is addition — all atoms of the monomer end up in the polymer.) The polyethylene has an average molecular weight of $280{,}000\ g/mol$. What is the average value of n? [Cards: 1, 13, 16]

C178. Reverse osmosis desalination: seawater ($0.60\ M\ NaCl$ at $298\ K$) is pressurized against a semipermeable membrane. (a) Calculate the minimum pressure (osmotic pressure) that must be overcome. ($R = 0.08206$, $i = 2$) (b) In practice, a pressure of $50\ atm$ is applied. What is the net driving pressure? (c) If the plant produces $1000\ m^3$ of fresh water per day, at an energy cost of $4.0\ kWh/m^3$, what is the daily energy consumption in GJ? ($1\ kWh = 3.6 \times 10^6\ J$) [Cards: 143, 144]

C179. Flue gas desulfurization: SO_2 is scrubbed from power plant exhaust with limestone slurry: $CaCO_3(s) + SO_2(g) + \frac{1}{2}O_2(g) \rightarrow CaSO_4(s) + CO_2(g)$. A plant emits $500\ kg$ of SO_2 ($M = 64.1$) per hour and must remove 95%. What mass of $CaCO_3$ ($M = 100.1$) is consumed per day? (24-hour operation) [Cards: 1, 8, 9, 10, 13]

C180. Cement production: the key reaction in a cement kiln is $CaCO_3(s) + SiO_2(s) \rightarrow CaSiO_3(s) + CO_2(g)$, $\Delta H^\circ = +88\ kJ/mol$. (a) $1000\ kg$ of $CaCO_3$ is heated with stoichiometric SiO_2. What mass of CO_2 is released? ($M_{CaCO_3} = 100.1$, $M_{CO_2} = 44.0$) (b) If the heat is supplied by burning coal (approximated as C → CO_2, $\Delta H^\circ = -393.5\ kJ/mol$) with 50% thermal efficiency, what mass of carbon is needed? ($M_C = 12.0$) (c) Cement production accounts for ~8% of global CO_2 emissions. Using your answer to (a) and (b), calculate the total CO_2 released (process + fuel) per ton of $CaCO_3$ processed. [Cards: 1, 8, 9, 10, 39, 40, 41, 44, 162]

SEMICONDUCTOR PROCESS CHEMISTRY — 30 Problems (Level 2–4, 2–7 cards)

> The cleanroom is a chemistry lab. Every chip layer — grown, deposited, etched, doped — is a chemical reaction controlled by the same cards you already know. The only new vocabulary is the process names. The equations are unchanged.

C181–C186: Silicon Purification & Crystal Growth

C181. Metallurgical-grade silicon is converted to trichlorosilane: $\mathrm{Si(s) + 3HCl(g) \rightarrow SiHCl_3(g) + H_2(g)}$. A reactor treats $100\ \mathrm{kg}$ of Si ($M = 28.1$) with excess HCl. What mass of $\mathrm{SiHCl_3}$ ($M = 135.5$) is produced at 90% yield? [Cards: 1, 8, 9, 10, 13]

C182. The Siemens process purifies Si by decomposing $\mathrm{SiHCl_3}$ onto a heated rod: $\mathrm{2SiHCl_3(g) + 2H_2(g) \rightarrow 2Si(s) + 6HCl(g)}$ (simplified). Or: $\mathrm{SiHCl_3(g) + H_2(g) \rightarrow Si(s) + 3HCl(g)}$. A Siemens reactor deposits $5.00\ \mathrm{kg}$ of ultra-pure Si ($M = 28.1$) per rod. What volume of HCl gas ($M = 36.5$) is produced at $900^\circ\mathrm{C}$ and $1.00\ \mathrm{atm}$ per rod? ($R = 0.08206$) [Cards: 1, 8, 9, 10, 19, 160]

C183. Silane ($\mathrm{SiH_4}$) is an alternative silicon precursor, decomposed by CVD: $\mathrm{SiH_4(g) \rightarrow Si(s) + 2H_2(g)}$. The decomposition is first-order with $k = 0.15\ \mathrm{s^{-1}}$ at $650^\circ\mathrm{C}$. What fraction of $\mathrm{SiH_4}$ decomposes in $10.0\ \mathrm{s}$? [Cards: 81, 83]

C184. Polysilicon production via the Siemens process operates at $1100^\circ\mathrm{C}$. At this temperature, the equilibrium constant for $\mathrm{SiHCl_3(g) + H_2(g) \rightleftharpoons Si(s) + 3HCl(g)}$ is $K = 0.85$ (in terms of gas-phase species). If a reactor initially contains $P_{\mathrm{SiHCl_3}} = 2.00\ \mathrm{atm}$ and $P_{\mathrm{H_2}} = 3.00\ \mathrm{atm}$ (with $P_{\mathrm{HCl}} = 0$), what is the equilibrium partial pressure of HCl? (Note: solid Si does not appear in K — Card 89.) The reaction stoichiometry gives the ICE changes: $SiHCl_3$: $-x$, H_2: $-x$, HCl: $+3x$. [Cards: 89, 93, 95]

C185. The Czochralski process melts polysilicon at $1414^\circ\mathrm{C}$. The latent heat of fusion of Si is $50.2\ \mathrm{kJ/mol}$. How much energy is required to melt $50.0\ \mathrm{kg}$ of Si ($M = 28.1$) starting from solid at $25^\circ\mathrm{C}$? The specific heat of solid Si is $0.705\ \mathrm{J\ g^{-1}\ K^{-1}}$. (Heat to melting point + latent heat.) [Cards: 1, 33, 39]

C186. During Czochralski crystal growth, a small amount of dopant (e.g., boron) is added to the melt. If $0.0100\ \mathrm{g}$ of boron ($M = 10.8$) is added to $50.0\ \mathrm{kg}$ of Si ($M = 28.1$), what is the boron concentration in (a) mass ppb, (b) mole fraction? [Cards: 17, 136, 138]

C187–C192: Thermal Oxidation & CVD

C187. Thermal oxidation of silicon: $\mathrm{Si(s) + O_2(g) \rightarrow SiO_2(s)}$. A $200\ \mathrm{mm}$ diameter wafer is oxidized until an $\mathrm{SiO_2}$ layer of thickness $100\ \mathrm{nm}$ ($1\ \mathrm{nm} = 10^{-9}\ \mathrm{m}$) grows. The density of $\mathrm{SiO_2}$ is $2.20\ \mathrm{g/cm^3}$, $M_{\mathrm{SiO_2}} = 60.1$. What mass of Si ($M = 28.1$) is consumed? (Volume of SiO_2 = wafer area × thickness. Wafer area = πr^2. The mole ratio Si:SiO_2 is 1:1.) [Cards: 1, 8, 9, 10, 158]

C188. Dry oxidation of Si follows the Deal-Grove model. In the linear regime (thin oxides), the growth rate is proportional to the $\mathrm{O_2}$ concentration at the surface. If the rate doubles when the $\mathrm{O_2}$ partial pressure is doubled (at constant T), what does this imply about the reaction order with respect to $\mathrm{O_2}$? Which card applies? [Cards: 76, 78]

C189. In a CVD reactor, $\mathrm{SiO_2}$ is deposited from $\mathrm{SiH_4}$ and $\mathrm{O_2}$: $\mathrm{SiH_4(g) + O_2(g) \rightarrow SiO_2(s) + 2H_2(g)}$. The reactor is at $400^\circ\mathrm{C}$ and $0.500\ \mathrm{torr}$ total pressure. $\mathrm{SiH_4}$ and $\mathrm{O_2}$ are fed in stoichiometric ratio. What volume of $\mathrm{SiH_4}$ gas (measured at STP) is consumed to deposit $1.00\ \mathrm{g}$ of $\mathrm{SiO_2}$ ($M = 60.1$)? [Cards: 1, 6, 7, 8, 9, 10, 160]

C190. Silicon nitride ($\mathrm{Si_3N_4}$) is deposited by LPCVD: $\mathrm{3SiH_2Cl_2(g) + 4NH_3(g) \rightarrow Si_3N_4(s) + 6HCl(g) + 6H_2(g)}$. (DCS = dichlorosilane, $\mathrm{SiH_2Cl_2}$.) A furnace processes 50 wafers simultaneously. Each wafer receives a $200\ \mathrm{nm}$ thick $\mathrm{Si_3N_4}$ film (density $3.44\ \mathrm{g/cm^3}$, $M_{\mathrm{Si_3N_4}} = 140.3$). Wafer diameter = $200\ \mathrm{mm}$. What total mass of $\mathrm{SiH_2Cl_2}$ ($M = 101.0$) is consumed? [Cards: 1, 8, 9, 10, 158]

C191. Tungsten CVD uses $\mathrm{WF_6}$ reduction by $\mathrm{H_2}$: $\mathrm{WF_6(g) + 3H_2(g) \rightarrow W(s) + 6HF(g)}$. The reaction is run at $400^\circ\mathrm{C}$ and $0.80\ \mathrm{atm}$ total pressure. If the feed gas is $20\%\ \mathrm{WF_6}$ and $80\%\ \mathrm{H_2}$ by volume, what is the partial pressure of HF after 50% of the $\mathrm{WF_6}$ has reacted? (Assume constant total pressure and temperature.) [Cards: 19, 26, 27]

C192. Atomic Layer Deposition (ALD) of $\mathrm{Al_2O_3}$ uses alternating pulses of trimethylaluminum ($\mathrm{Al(CH_3)_3}$) and water: $\mathrm{2Al(CH_3)_3(g) + 3H_2O(g) \rightarrow Al_2O_3(s) + 6CH_4(g)}$. Each ALD cycle deposits $0.10\ \mathrm{nm}$ of $\mathrm{Al_2O_3}$ (density $3.95\ \mathrm{g/cm^3}$, $M_{\mathrm{Al_2O_3}} = 102.0$). How many cycles are needed to deposit a $10.0\ \mathrm{nm}$ film? What mass of $\mathrm{Al(CH_3)_3}$ ($M = 72.1$) is consumed per $\mathrm{cm^2}$ of wafer area for the full $10.0\ \mathrm{nm}$ film? [Cards: 1, 8, 9, 10, 158]

C193–C198: Etching & Cleaning

C193. Wet etching of $\mathrm{SiO_2}$ uses buffered HF: $\mathrm{SiO_2(s) + 6HF(aq) \rightarrow H_2SiF_6(aq) + 2H_2O(l)}$. In a semiconductor fab, a $200\ \mathrm{mm}$ wafer with a $500\ \mathrm{nm}$ thick $\mathrm{SiO_2}$ layer (density $2.20\ \mathrm{g/cm^3}$) is completely etched. What volume of 49% HF solution (by mass, density $1.18\ \mathrm{g/mL}$) is required? ($M_{\mathrm{HF}} = 20.0$, $M_{\mathrm{SiO_2}} = 60.1$) [Cards: 1, 8, 9, 10, 133, 135, 158]

C194. Plasma etching of silicon uses $\mathrm{SF_6}$: $\mathrm{2SF_6(g) \rightarrow 2SF_4(g) + 2F(g)}$ (simplified; F atoms then etch Si: $\mathrm{Si + 4F \rightarrow SiF_4}$). A plasma etcher consumes $0.500\ \mathrm{g}$ of Si ($M = 28.1$) per minute from a wafer. What mass of $\mathrm{SF_6}$ ($M = 146.1$) must be supplied per hour? (Assume 100% F utilization.) [Cards: 1, 8, 9, 10]

C195. In reactive ion etching (RIE), the etch rate of $\mathrm{SiO_2}$ in a $\mathrm{CF_4/O_2}$ plasma is $50\ \mathrm{nm/min}$. The selectivity (ratio of etch rates) of $\mathrm{SiO_2}$ to Si is $20:1$. After etching through a $200\ \mathrm{nm}\ \mathrm{SiO_2}$ layer, how much underlying Si (in nm) is unintentionally etched during a 20% over-etch (etching continues for 20% longer than the calculated endpoint time)? [Cards: — (arithmetic/applied)]

C196. RCA clean Step 1 (Standard Clean 1) uses $\mathrm{NH_4OH/H_2O_2/H_2O}$ ($1:1:5$ ratio) at $75^\circ\mathrm{C}$ to remove organic contaminants. The solution is prepared by mixing $100\ \mathrm{mL}$ of $29\%\ \mathrm{NH_3}$ (density $0.90\ \mathrm{g/mL}$, $M_{\mathrm{NH_3}} = 17.0$), $100\ \mathrm{mL}$ of $30\%\ \mathrm{H_2O_2}$ (density $1.11\ \mathrm{g/mL}$, $M_{\mathrm{H_2O_2}} = 34.0$), and $500\ \mathrm{mL}$ of water. What is the molarity of $\mathrm{NH_3}$ and $\mathrm{H_2O_2}$ in the final solution? [Cards: 133, 137]

C197. After RCA cleaning, wafers are rinsed in deionized water until the resistivity of the rinse water reaches $18.2\ \mathrm{M\Omega\ cm}$ (the theoretical limit for pure water). The conductivity κ is the reciprocal of resistivity: $\kappa = 1/\rho$. For pure water, $\kappa_w = 0.055\ \mu\mathrm{S/cm}$ ($\mathrm{S} = \Omega^{-1}$). From $K_w = 1.0 \times 10^{-14}$, calculate the theoretical $[\mathrm{H^+}]$ and $[\mathrm{OH^-}]$ in pure water at $25^\circ\mathrm{C}$, and verify that pure water has essentially zero ionic contaminants. [Cards: 101, 103]

C198. Piranha etch ($\mathrm{H_2SO_4/H_2O_2}$, $3:1$) cleans organic residues from wafers at $120^\circ\mathrm{C}$. The solution is prepared by carefully adding $300\ \mathrm{mL}$ of $98\%\ \mathrm{H_2SO_4}$ (density $1.84\ \mathrm{g/mL}$, $M = 98.1$) to $100\ \mathrm{mL}$ of $30\%\ \mathrm{H_2O_2}$ (density $1.11\ \mathrm{g/mL}$, $M = 34.0$). What is the mole fraction of $\mathrm{H_2SO_4}$ in the final mixture? [Cards: 136]

C199–C204: Doping, Diffusion & Photolithography

C199. Ion implantation: $\mathrm{As^+}$ ions are accelerated through $50\ \mathrm{keV}$ and implanted into Si. The dose is $1.0 \times 10^{15}\ \mathrm{ions/cm^2}$ over a $200\ \mathrm{mm}$ wafer. How many grams of As ($M = 74.9$) are implanted per wafer? ($N_A = 6.022 \times 10^{23}$) [Cards: 4, 5]

C200. After implantation, the dopant is activated by rapid thermal annealing at $1000°C$ for 30 s. The diffusion coefficient of boron in Si at $1000°C$ is $D = 2.0 \times 10^{-14}\ cm^2/s$. Using the diffusion length formula $L \approx \sqrt{Dt}$, estimate how far (in nm) boron atoms diffuse during the anneal. ($1\ nm = 10^{-7}\ cm$) [Cards: 30]

C201. The diffusion of phosphorus in Si follows an Arrhenius temperature dependence: $D = D_0 e^{-E_a/(RT)}$ with $D_0 = 3.85\ cm^2/s$ and $E_a = 3.66\ eV/atom$. ($1\ eV = 96.5\ kJ/mol$. Use $R = 8.314\ J\ mol^{-1}\ K^{-1}$, and convert E_a to J/mol.) Calculate D at $1000°C$. [Cards: 84, 85, 205]

C202. Photolithography uses a photoresist that reacts according to first-order kinetics under UV exposure. The exposure dose D (mJ/cm²) is related to the fraction f of unreacted resist by $f = e^{-kD}$, where $k = 0.030\ cm^2/mJ$. What dose is required for 99% of the resist to react ($f = 0.01$)? [Cards: 81]

C203. The photoresist from C202 has a contrast $\gamma = 3.0$, defined by $\gamma = \frac{1}{\log_{10}(D_0/D_1)}$ where D_0 is the dose to just clear the resist and D_1 is the dose where the resist just begins to clear. If $D_1 = 50\ mJ/cm^2$, what is D_0? [Cards: — (applied logarithm)]

C204. Developer solution for photoresist is typically 2.38% tetramethylammonium hydroxide ($TMAH$, $N(CH_3)_4OH$, $M = 91.2$) by mass. A fab uses $500\ L$ of developer per day. The concentrated TMAH solution is 25% by mass (density $1.00\ g/mL$). What volume of concentrated TMAH is needed to prepare one day's supply of developer? (Assume developer density ≈ $1.00\ g/mL$.) [Cards: 133, 135, 137]

C205–C210: Advanced Device Fabrication & Materials

C205. Copper electroplating for damascene interconnects uses $CuSO_4/H_2SO_4$ bath. The half-reaction is $Cu^{2+} + 2e^- \rightarrow Cu$. A $200\ mm$ wafer has 10^4 trenches, each $0.10\ \mu m$ wide, $0.30\ \mu m$ deep, and $1.0\ cm$ long. What total charge (in coulombs) is required to fill all trenches with Cu (density $8.96\ g/cm^3$, $M_{Cu} = 63.5$)? ($F = 96{,}485$) [Cards: 1, 129, 131, 132, 158]

C206. Chemical Mechanical Polishing (CMP) removes excess Cu after electroplating. The CMP slurry contains $5.0\%\ H_2O_2$ by mass (density $1.02\ g/mL$, $M_{H_2O_2} = 34.0$) to oxidize the Cu surface: $Cu + H_2O_2 \rightarrow CuO + H_2O$. If the slurry flow rate is $200\ mL/min$, how many moles of Cu can be oxidized per minute? (Assume H_2O_2 is the limiting reagent at the wafer surface.) [Cards: 1, 8, 9, 11, 133, 135]

C207. Hafnium oxide (HfO_2) is a high-κ gate dielectric deposited by ALD: $HfCl_4(g) + 2H_2O(g) \rightarrow HfO_2(s) + 4HCl(g)$. The deposited film is $2.0\ nm$ thick on a $300\ mm$ wafer. Density of HfO_2 is $9.68\ g/cm^3$, $M_{HfO_2} = 210.5$. (a) What mass of HfO_2 is on the wafer? (b) What mass of $HfCl_4$ ($M = 320.3$) was consumed? [Cards: 1, 8, 9, 10, 158]

C208. The gate dielectric in C207 requires an equivalent oxide thickness (EOT) of $0.50\ nm$. EOT relates the physical thickness $t_{\text{high-}\kappa}$ to the equivalent SiO_2 thickness: $EOT = t_{\text{high-}\kappa} \times \frac{\kappa_{SiO_2}}{\kappa_{\text{high-}\kappa}}$, where $\kappa_{SiO_2} = 3.9$ and $\kappa_{HfO_2} = 25$. What physical thickness of HfO_2 is needed to achieve this EOT? Does the $2.0\ nm$ film from C207 meet the requirement? [Cards: — (applied)]

C209. Titanium nitride (TiN) is deposited as a diffusion barrier by reactive sputtering: $Ti(s) + \frac{1}{2}N_2(g) \rightarrow TiN(s)$, $\Delta H° = -338\ kJ/mol$. A sputtering target contains $5.00\ kg$ of Ti ($M = 47.9$). (a) How many moles of N_2 gas are consumed to fully react the Ti target? (b) What volume of N_2 at STP is required? ($M_{N_2} = 28.0$) [Cards: 1, 6, 7, 8, 9, 39, 40]

C210. A state-of-the-art logic chip contains ~5.0×10^{10} transistors on a $300\ \mathrm{mm}$ wafer. Each transistor's gate is a fin of Si ($0.020\ \mu\mathrm{m}$ wide × $0.050\ \mu\mathrm{m}$ tall × $0.10\ \mu\mathrm{m}$ long, density of Si = $2.33\ \mathrm{g/cm^3}$). (a) What is the total mass of Si in all the transistor fins on one wafer? (b) This Si was deposited epitaxially from $\mathrm{SiH_4}$ (C183). What mass of $\mathrm{SiH_4}$ ($M = 32.1$) was required, assuming 100% incorporation? (c) The fab processes $10{,}000$ wafers per month. What is the annual $\mathrm{SiH_4}$ consumption in kg? [Cards: 1, 4, 5, 8, 9, 10, 158]

ULTRA-INDUSTRIAL — 10 Problems (Level 5, 7–10 domains)

> These are not "problems" in the ordinary sense. They are miniature chemical engineering design exercises. Each one chains 7 or more domains from the vocabulary deck. The Trigger Trace for a single problem may fire 12 cards across stoichiometry, gases, thermochemistry, kinetics, equilibrium, electrochemistry, solutions, and bridges. Solve one per day. They are meant to be hard.

C211–C215: Integrated Process Design — Full Plant Optimization

C211. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Bridges] A Haber-Bosch reactor operates at $450^\circ\mathrm{C}$ and $250\ \mathrm{atm}$. The feed is a $3:1\ \mathrm{H_2:N_2}$ mixture at $25^\circ\mathrm{C}$. (a) The feed must be preheated to $450^\circ\mathrm{C}$ before entering the reactor. The average C_p of the $3:1\ \mathrm{H_2:N_2}$ mixture is $29.1\ \mathrm{J\ mol^{-1}\ K^{-1}}$. How much heat (kJ) is required to preheat $1000\ \mathrm{mol}$ of feed gas? (b) At $450^\circ\mathrm{C}$, $K_p = 4.3 \times 10^{-5}\ \mathrm{atm^{-2}}$ for $\mathrm{N_2 + 3H_2 \rightleftharpoons 2NH_3}$. The reactor achieves 85% of the equilibrium-limited conversion. Using an ICE table with initial $P_{\mathrm{N_2}} = 62.5\ \mathrm{atm}$ and $P_{\mathrm{H_2}} = 187.5\ \mathrm{atm}$, find the equilibrium $P_{\mathrm{NH_3}}$, then the actual $P_{\mathrm{NH_3}}$ at 85% of that. (c) The exothermic reaction ($\Delta H^\circ = -92.2\ \mathrm{kJ/mol}$) heats the product stream. If $15\ \mathrm{mol}$ of $\mathrm{NH_3}$ are produced, how much heat is released? (d) This heat is recovered to generate steam. Water enters the heat exchanger at $25^\circ\mathrm{C}$ and exits as steam at $200^\circ\mathrm{C}$ ($c_{p,\mathrm{water}} = 4.18\ \mathrm{J\ g^{-1}\ K^{-1}}$, $c_{p,\mathrm{steam}} = 2.00\ \mathrm{J\ g^{-1}\ K^{-1}}$, $\Delta H_{\mathrm{vap}} = 40.7\ \mathrm{kJ/mol}$ at $100^\circ\mathrm{C}$, $M_{\mathrm{H_2O}} = 18.0$). How many kg of steam can be generated from the heat in (c)? (e) The ammonia is condensed out at $-33^\circ\mathrm{C}$ ($\Delta H_{\mathrm{vap,NH_3}} = 23.3\ \mathrm{kJ/mol}$, $M_{\mathrm{NH_3}} = 17.0$). How much additional cooling (kJ) is required to condense the $15\ \mathrm{mol}$ of $\mathrm{NH_3}$? (f) Unreacted $\mathrm{N_2}$ and $\mathrm{H_2}$ are recycled. If the single-pass conversion of $\mathrm{N_2}$ is 12%, use first-order recycle mathematics to find how many passes achieve 95% overall conversion. [Cards: 1, 8, 9, 19, 21, 26, 27, 33, 34, 38, 39, 40, 41, 48, 81, 89, 93, 95, 99, 158, 160, 162, 197, 199]

C212. [8 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Electrochem + Solutions + Bridges] An integrated methanol plant: Step 1 — Steam reforming: $\mathrm{CH_4 + H_2O \rightleftharpoons CO + 3H_2}$ ($\Delta H^\circ = +206\ \mathrm{kJ/mol}$) at $850^\circ\mathrm{C}$, $25\ \mathrm{atm}$. Step 2 — Water-gas shift: $\mathrm{CO + H_2O \rightleftharpoons CO_2 + H_2}$ ($\Delta H^\circ = -41.2\ \mathrm{kJ/mol}$) at $250^\circ\mathrm{C}$. Step 3 — Methanol synthesis: $\mathrm{CO + 2H_2 \rightleftharpoons CH_3OH}$ ($\Delta H^\circ = -90.7\ \mathrm{kJ/mol}$) at $250^\circ\mathrm{C}$, $75\ \mathrm{atm}$. (a) In Step 1, $1000\ \mathrm{mol\ CH_4}$ and $3000\ \mathrm{mol\ H_2O}$ are fed. The equilibrium constant at $850^\circ\mathrm{C}$ is $K_p = 1.2 \times 10^4\ \mathrm{atm^2}$. If 90% of $\mathrm{CH_4}$ reacts, how many moles of $\mathrm{H_2}$ exit the reformer? (b) The heat for the endothermic reforming is supplied by burning part of the $\mathrm{CH_4}$ feed: $\mathrm{CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O}$, $\Delta H^\circ = -890\ \mathrm{kJ/mol}$. How many moles of $\mathrm{CH_4}$ must be burned to supply the heat for Step 1, assuming 70% thermal efficiency? (c) In Step 2, the shift reactor at $250^\circ\mathrm{C}$ has $K_p = 2.8 \times 10^2$. With the outlet from (a), calculate Q and determine the equilibrium composition. (d) Before Step 3, the gas is compressed from 25 to $75\ \mathrm{atm}$. Using $P_1V_1 = P_2V_2$ (isothermal compression at $250^\circ\mathrm{C}$), by what factor does the volume decrease? (e) In Step 3, the methanol reactor at $250^\circ\mathrm{C}$ and $75\ \mathrm{atm}$ has $K_p = 2.5 \times 10^{-2}\ \mathrm{atm^{-2}}$. If the inlet $\mathrm{CO : H_2}$ ratio is $1:2$ and the total inlet pressure of reactants is

$60\ \mathrm{atm}$, set up the ICE table. (f) The crude methanol is purified by distillation. The methanol-water mixture (30% water by mass) is distilled. How much heat is needed to vaporize $100\ \mathrm{kg}$ of the mixture? ($\Delta H_{\mathrm{vap,CH_3OH}} = 35.2\ \mathrm{kJ/mol}$, $M = 32.0$; $\Delta H_{\mathrm{vap,H_2O}} = 40.7\ \mathrm{kJ/mol}$, $M = 18.0$) [Cards: 1, 8, 9, 10, 11, 19, 21, 26, 27, 33, 38, 39, 40, 41, 42, 48, 89, 90, 91, 93, 95, 135, 160, 162]

C213. [7 domains: Stoich + Gases + Thermo + Electrochem + Kinetics + Equilibrium + Bridges] The chlor-alkali membrane cell produces $\mathrm{Cl_2}$, $\mathrm{H_2}$, and NaOH. Downstream, $\mathrm{Cl_2}$ is reacted with $\mathrm{H_2}$ to make HCl: $\mathrm{H_2 + Cl_2 \rightarrow 2HCl}$, $\Delta H^\circ = -184.6\ \mathrm{kJ/mol}$. The HCl is then oxidized back to $\mathrm{Cl_2}$ via the Deacon process: $\mathrm{4HCl + O_2 \rightleftharpoons 2Cl_2 + 2H_2O}$, $\Delta H^\circ = -114.4\ \mathrm{kJ/mol}$, catalyzed by $\mathrm{CuCl_2}$ at $450^\circ\mathrm{C}$. (a) A chlor-alkali cell runs at $200{,}000\ \mathrm{A}$ with 95% current efficiency. How many kg of $\mathrm{Cl_2}$ ($M = 71.0$) are produced per day? ($F = 96{,}485$) The half-reaction: $\mathrm{2Cl^- \rightarrow Cl_2 + 2e^-}$. (b) The $\mathrm{Cl_2}$ from (a) is fully converted to HCl. What mass of $\mathrm{H_2}$ ($M = 2.02$) from the same cell is consumed? (Half-reaction: $\mathrm{2H_2O + 2e^- \rightarrow H_2 + 2OH^-}$ — the $\mathrm{H_2}$ and $\mathrm{Cl_2}$ are produced in a 1:1 mole ratio.) (c) The HCl is oxidized via the Deacon process. At $450^\circ\mathrm{C}$, $K_p = 58\ \mathrm{atm^{-1}}$ for $\mathrm{4HCl + O_2 \rightleftharpoons 2Cl_2 + 2H_2O}$. The feed is stoichiometric HCl and $\mathrm{O_2}$ at $1.50\ \mathrm{atm}$ total initial pressure. Set up the ICE table and find the equilibrium conversion of HCl. (d) The Deacon process is exothermic. The reactor operates adiabatically and the temperature rises. The rate constant k follows Arrhenius with $E_a = 85\ \mathrm{kJ/mol}$. If the inlet is at $450^\circ\mathrm{C}$ and the outlet reaches $520^\circ\mathrm{C}$, by what factor does k increase? ($R = 8.314$) (e) Calculate ΔG° for the Deacon process at $450^\circ\mathrm{C}$ from K_p and explain why a catalyst is essential despite favorable thermodynamics. [Cards: 1, 8, 9, 10, 39, 40, 41, 48, 49, 51, 84, 85, 86, 88, 89, 93, 95, 121, 122, 125, 126, 129, 131, 132, 164, 165, 166]

C214. [8 domains: Stoich + Gases + Thermo + Acid-Base + Equilibrium + Kinetics + Solutions + Bridges] Ostwald process for nitric acid: Step 1 — $\mathrm{4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O}$ (Pt/Rh catalyst, $900^\circ\mathrm{C}$, $\Delta H^\circ = -905\ \mathrm{kJ/mol}$). Step 2 — $\mathrm{2NO + O_2 \rightarrow 2NO_2}$ ($\Delta H^\circ = -114\ \mathrm{kJ/mol}$, homogeneous, third-order). Step 3 — $\mathrm{3NO_2 + H_2O \rightarrow 2HNO_3 + NO}$ (absorption tower). (a) Step 1: $500\ \mathrm{kg}$ of $\mathrm{NH_3}$ ($M = 17.0$) is oxidized per hour at 95% conversion. What mass of NO ($M = 30.0$) is produced per hour? (b) The oxidation of NO to $\mathrm{NO_2}$ (Step 2) is third-order: Rate = $k[\mathrm{NO}]^2[\mathrm{O_2}]$ with $k = 7.1 \times 10^3\ \mathrm{M^{-2}\ s^{-1}}$ at $298\ \mathrm{K}$. The reaction is run at $50^\circ\mathrm{C}$ where k is much slower (it actually decreases with T for this unusual reaction — E_a is negative: $-4.6\ \mathrm{kJ/mol}$). At $50^\circ\mathrm{C}$, $k = 4.2 \times 10^3\ \mathrm{M^{-2}\ s^{-1}}$. If $[\mathrm{NO}] = 0.010\ \mathrm{M}$ and $[\mathrm{O_2}] = 0.0050\ \mathrm{M}$, what is the initial rate at $50^\circ\mathrm{C}$? (c) The $\mathrm{NO_2}$ absorption (Step 3) produces $60\%\ \mathrm{HNO_3}$ by mass. A storage tank holds $10{,}000\ \mathrm{kg}$ of this acid. The density of $60\%\ \mathrm{HNO_3}$ is $1.37\ \mathrm{g/mL}$, $M_{\mathrm{HNO_3}} = 63.0$. What is the molarity of $\mathrm{HNO_3}$? (d) The tail gas contains unabsorbed NO. This is removed by reacting with $\mathrm{NH_3}$: $\mathrm{4NO + 4NH_3 + O_2 \rightarrow 4N_2 + 6H_2O}$ (selective catalytic reduction). If the tail gas contains $500\ \mathrm{ppm}$ NO by volume at $1.00\ \mathrm{atm}$ and $200^\circ\mathrm{C}$, what is the partial pressure of NO in atm? ($1\ \mathrm{ppm} = 10^{-6}$ by volume.) What mass of $\mathrm{NH_3}$ is needed per $1000\ \mathrm{m^3}$ of tail gas (measured at $200^\circ\mathrm{C}$, $1.00\ \mathrm{atm}$)? ($R = 0.08206$) (e) The nitric acid is neutralized with $\mathrm{NH_3}$ to make ammonium nitrate fertilizer: $\mathrm{HNO_3 + NH_3 \rightarrow NH_4NO_3}$. If all the acid from (a) is neutralized, what mass of $\mathrm{NH_4NO_3}$ ($M = 80.0$) is produced per hour? [Cards: 1, 8, 9, 10, 11, 13, 19, 20, 26, 33, 39, 40, 76, 77, 78, 85, 86, 101, 107, 133, 134, 135, 160]

C215. [7 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Solutions + Bridges] Integrated Gasification Combined Cycle (IGCC) with pre-combustion $\mathrm{CO_2}$ capture: Coal ($\approx$ C) is gasified: $\mathrm{C + H_2O \rightarrow CO + H_2}$ ($\Delta H^\circ = +131\ \mathrm{kJ/mol}$) at $1200^\circ\mathrm{C}$, $30\ \mathrm{atm}$. The syngas is shifted: $\mathrm{CO + H_2O \rightleftharpoons CO_2 + H_2}$ ($\Delta H^\circ = -41.2\ \mathrm{kJ/mol}$). $\mathrm{CO_2}$ is removed by Selexol (physical solvent, absorbs $\mathrm{CO_2}$ at high pressure). The remaining $\mathrm{H_2}$ is burned in a gas turbine. (a) A gasifier processes $1000\ \mathrm{kg}$ of carbon ($M = 12.0$) per hour. How many moles of syngas ($\mathrm{CO + H_2}$) are produced per hour, assuming complete conversion? (b) The syngas enters the shift reactor at $400^\circ\mathrm{C}$ and $30\ \mathrm{atm}$. $K_p = 11.7$ at

$400°C$. Initially, the syngas from (a) has $CO : H_2O : H_2 : CO_2 = 1 : 3 : 1 : 0$ mole ratio (steam is added before shifting). Set up the ICE table and find the equilibrium composition. (c) CO_2 is removed with 95% efficiency by Selexol absorption. The absorbed CO_2 is released when the pressure is dropped to $1\ \mathrm{atm}$. What volume of CO_2 (at $25°C, 1\ \mathrm{atm}$) is captured per hour? (d) The remaining H_2-rich gas is burned: $2H_2 + O_2 \rightarrow 2H_2O$, $\Delta H° = -571.6\ \mathrm{kJ/mol}$. Calculate the hourly heat output in GJ. (e) The combustion produces steam at $600°C$ and $100\ \mathrm{atm}$ that drives a turbine. The steam's C_p is $2.10\ \mathrm{J\ g^{-1}\ K^{-1}}$. If the steam cools to $100°C$ in the turbine, how much energy is extracted per kg of steam? If the turbine-generator is 40% efficient, what is the electrical output in MW? ($1\ \mathrm{MW} = 10^6\ \mathrm{J/s}$) (f) The plant must meet a CO_2 emission limit of $100\ \mathrm{kg\ CO_2/MWh}$. With the 95% capture from (c), does it comply? Calculate $\mathrm{kg\ CO_2}$ emitted per MWh of electricity from (e). [Cards: 1, 8, 9, 10, 19, 20, 26, 27, 33, 34, 38, 39, 40, 41, 48, 89, 90, 93, 95, 96, 99, 160, 162, 172, 197]

C216–C220: Advanced Industrial Synthesis & Environmental Control

C216. [7 domains: Stoich + Gases + Thermo + Equilibrium + Acid-Base + Solutions + Bridges] Flue gas desulfurization (FGD) by the limestone-gypsum wet scrubbing process: SO_2 is absorbed into a limestone slurry, oxidized, and crystallized as gypsum. $SO_2(g) \rightleftharpoons SO_2(aq)$, Henry's Law constant $K_H = 1.2\ \mathrm{M/atm}$ at $55°C$. $SO_2(aq) + H_2O \rightleftharpoons H^+ + HSO_3^-$, $K_{a1} = 1.7 \times 10^{-2}$. $CaCO_3(s) + 2H^+ \rightarrow Ca^{2+} + CO_2 + H_2O$. $HSO_3^- + \frac{1}{2}O_2 \rightarrow SO_4^{2-} + H^+$. $Ca^{2+} + SO_4^{2-} + 2H_2O \rightarrow CaSO_4 \cdot 2H_2O(s)$ (gypsum). (a) A power plant exhausts $10^6\ \mathrm{m^3/hr}$ of flue gas at $55°C$ and $1.00\ \mathrm{atm}$ containing $1500\ \mathrm{ppm}\ SO_2$ by volume. What is the partial pressure of SO_2 in atm? ($1\ \mathrm{ppm} = 10^{-6}$) (b) Using Henry's Law, what is the equilibrium concentration of dissolved SO_2 at the gas-liquid interface? (c) The dissolved SO_2 dissociates into HSO_3^- (first dissociation dominates). From K_{a1}, calculate the equilibrium $[H^+]$ and pH at the interface, assuming $[SO_2(aq)] = 1.8 \times 10^{-3}\ \mathrm{M}$ from (b). (d) $CaCO_3$ is added to neutralize the acid. What is the minimum mass of $CaCO_3$ ($M = 100.1$) required per hour to neutralize the H^+ produced, assuming all SO_2 ultimately produces $2\ H^+$ per SO_2 (from the overall stoichiometry: $SO_2 + \frac{1}{2}O_2 + CaCO_3 + 2H_2O \rightarrow CaSO_4 \cdot 2H_2O + CO_2$)? (e) The gypsum produced is $CaSO_4 \cdot 2H_2O$ ($M = 172.2$). What is the daily gypsum production in metric tons? (f) The scrubbing liquor is recycled. Lime is sometimes added to regenerate the limestone: $CaO + H_2O \rightarrow Ca(OH)_2$, $\Delta H° = -65.2\ \mathrm{kJ/mol}$. How much heat is released per ton of CaO ($M = 56.1$) slaked? [Cards: 1, 8, 9, 10, 11, 19, 26, 33, 39, 40, 41, 89, 93, 95, 101, 104, 107, 109, 133, 160]

C217. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Bridges] Fischer-Tropsch (FT) synthesis converts syngas to liquid hydrocarbons: $CO + 2H_2 \rightarrow –CH_2– + H_2O$ (simplified), $\Delta H° \approx -165\ \mathrm{kJ/mol\ CO}$. A cobalt catalyst at $220°C$ and $25\ \mathrm{atm}$ gives C_{5+} selectivity of 85%. (a) A FT reactor receives $5000\ \mathrm{mol/hr}$ of syngas with $H_2/CO = 2.0$. At 60% single-pass CO conversion, how many moles of H_2O are produced per hour? (b) The FT reaction is highly exothermic. The reactor tubes are cooled by boiling water at $250°C$ ($\Delta H_{\mathrm{vap}} = 40.7\ \mathrm{kJ/mol}$ modified for pressure). If the heat released is $165\ \mathrm{kJ}$ per mole of CO converted, how many kg of steam are generated per hour? ($M_{H_2O} = 18.0$) (c) The FT wax (C_{20+}) is hydrocracked to diesel: $C_{20}H_{42} + H_2 \rightarrow C_{10}H_{22} + C_{10}H_{22}$ (simplified). If $100\ \mathrm{kg}$ of wax ($M \approx 282$) yields $85\ \mathrm{kg}$ of diesel ($M \approx 142$), what is the percent yield? (d) The aqueous phase from FT contains 2.0% methanol and 1.0% ethanol by mass (density $1.00\ \mathrm{g/mL}$). What is the molarity of methanol ($M = 32.0$) in this stream? (e) Unconverted syngas is recycled. If the recycle ratio (recycle flow / fresh feed flow) is 3.0, and the single-pass CO conversion is 60%, what is the overall CO conversion? (Use: overall = single-pass / [1 – (1 – single-pass) × (recycle fraction)] where recycle fraction = recycle ratio/(1 + recycle ratio).) (f) The tail gas ($CO + H_2 + CH_4 + CO_2$) is burned in a flare. The tail gas has an average heating value of $250\ \mathrm{kJ/mol}$. If $100\ \mathrm{mol/hr}$ of tail gas is flared, what is the heat output in MJ/hr, and what mass of CO_2 ($M = 44.0$) is emitted if the average carbon number of the tail gas is 0.3 mol C per mol gas? [Cards: 1, 8, 9, 10, 13, 19, 33, 38, 39, 40, 41, 81, 89, 93, 95, 133, 134, 158, 160]

C218. [8 domains: Stoich + Gases + Thermo + Electrochem + Kinetics + Equilibrium + Solutions + Bridges] Hydrogen economy: green $\mathrm{H_2}$ from PEM electrolysis → compressed storage → fuel cell vehicle. (a) A PEM electrolyzer operates at $1.80\ \mathrm{V}$ and $500{,}000\ \mathrm{A}$. How many kg of $\mathrm{H_2}$ ($M = 2.02$) are produced per hour? ($F = 96{,}485$, half-reaction: $2\mathrm{H^+} + 2\mathrm{e^-} \rightarrow \mathrm{H_2}$) (b) The electrolyzer consumes $1.80\ \mathrm{V} \times 500{,}000\ \mathrm{A} = 900\ \mathrm{kW}$ of electrical power. The higher heating value (HHV) of $\mathrm{H_2}$ is $286\ \mathrm{kJ/mol}$. What is the energy efficiency (HHV of $\mathrm{H_2}$ produced / electrical energy input)? (c) The $\mathrm{H_2}$ is compressed from $1\ \mathrm{atm}$ to $700\ \mathrm{atm}$ at $25^\circ\mathrm{C}$ for storage. Using $P_1V_1 = P_2V_2$, what volume does $1.00\ \mathrm{kg}$ of $\mathrm{H_2}$ occupy at $700\ \mathrm{atm}$? (At $1\ \mathrm{atm}$, $25^\circ\mathrm{C}$, $n = 1000/2.02 = 495\ \mathrm{mol}$, $V = nRT/P$.) Is the ideal gas assumption reasonable at $700\ \mathrm{atm}$? Name the card that corrects for this (van der Waals). (d) The fuel cell: $2\mathrm{H_2} + \mathrm{O_2} \rightarrow 2\mathrm{H_2O}$, $E^\circ = 1.23\ \mathrm{V}$. At an operating current of $200\ \mathrm{A}$ and voltage of $0.70\ \mathrm{V}$, what is the fuel cell's voltage efficiency? ($\eta_V = V_{\text{operating}}/E^\circ$) (e) At $200\ \mathrm{A}$, how many grams of $\mathrm{H_2}$ are consumed per hour? (f) The fuel cell operates at $80^\circ\mathrm{C}$. The Nernst equation predicts the thermodynamic potential at this temperature and at $P_{\mathrm{H_2}} = P_{\mathrm{O_2}} = 2.0\ \mathrm{atm}$ (product water is liquid). Calculate the Nernst-adjusted potential: $E = E^\circ - \dfrac{RT}{nF}\ln\dfrac{1}{P_{\mathrm{H_2}}P_{\mathrm{O_2}}^{1/2}}$. ($R = 8.314, T = 353\ \mathrm{K}, n = 2$) How does this compare to the actual operating voltage of $0.70\ \mathrm{V}$? What causes the difference? [Cards: 1, 19, 20, 32, 39, 40, 48, 121, 122, 125, 126, 127, 128, 129, 131, 132, 160, 197]

C219. [7 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Solutions + Bridges] Urea production: $2\mathrm{NH_3} + \mathrm{CO_2} \rightleftharpoons \mathrm{NH_2CONH_2} + \mathrm{H_2O}$ ($\Delta H^\circ = -117\ \mathrm{kJ/mol}$). The reaction is run at $185^\circ\mathrm{C}$ and $180\ \mathrm{atm}$ with excess $\mathrm{NH_3}$. (a) A urea plant receives $500\ \mathrm{kg/hr}$ of $\mathrm{NH_3}$ ($M = 17.0$) and $650\ \mathrm{kg/hr}$ of $\mathrm{CO_2}$ ($M = 44.0$). Which is the limiting reagent? What is the theoretical urea production rate in kg/hr? ($M_{\text{urea}} = 60.1$) (b) The equilibrium constant at $185^\circ\mathrm{C}$ is $K = 1.5$ (based on molar concentrations in the liquid phase). The reactor achieves 75% of equilibrium conversion of the limiting reagent. What is the actual urea production rate? (c) Unreacted $\mathrm{NH_3}$ and $\mathrm{CO_2}$ are stripped and recycled. The stripper uses heat: incoming solution at $170^\circ\mathrm{C}$ must be heated to $200^\circ\mathrm{C}$. The solution's C_p is $3.5\ \mathrm{J\ g^{-1}\ K^{-1}}$ and the flow rate is $2000\ \mathrm{kg/hr}$. Heat is supplied by steam at $220^\circ\mathrm{C}$ ($\Delta H_{\text{vap}} = 38.0\ \mathrm{kJ/mol}$ at pressure). How many kg of steam are needed per hour? ($M_{\mathrm{H_2O}} = 18.0$) (d) The urea melt is prilled (sprayed into droplets that solidify). The melt enters at $140^\circ\mathrm{C}$ and cools to $80^\circ\mathrm{C}$. The specific heat of urea melt is $2.0\ \mathrm{J\ g^{-1}\ K^{-1}}$, and the latent heat of solidification is $14.5\ \mathrm{kJ/mol}$ ($M_{\text{urea}} = 60.1$). For $1000\ \mathrm{kg/hr}$ of urea, what is the total heat that must be removed? (e) The prilling tower uses air at $25^\circ\mathrm{C}$. The air exits at $60^\circ\mathrm{C}$. Air's C_p is $1.01\ \mathrm{J\ g^{-1}\ K^{-1}}$. What mass of cooling air is needed per hour? ($\rho_{\text{air}} \approx 1.2\ \mathrm{kg/m^3}$ at $25^\circ\mathrm{C}$. What volume of air is this?) [Cards: 1, 8, 9, 10, 11, 12, 13, 14, 19, 33, 34, 38, 39, 40, 89, 93, 95, 135, 160]

C220. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Electrochem + Solutions] Titanium dioxide ($\mathrm{TiO_2}$) pigment via the Chloride Process: Step 1 — Carbochlorination: $\mathrm{TiO_2(s)} + 2\mathrm{Cl_2(g)} + \mathrm{C(s)} \rightarrow \mathrm{TiCl_4(g)} + \mathrm{CO_2(g)}$ at $900^\circ\mathrm{C}$, $\Delta H^\circ = -233\ \mathrm{kJ/mol}$. Step 2 — Purification: $\mathrm{TiCl_4}$ is distilled (bp $136^\circ\mathrm{C}$, $\Delta H_{\text{vap}} = 36.0\ \mathrm{kJ/mol}$, $M = 189.7$). Step 3 — Oxidation: $\mathrm{TiCl_4(g)} + \mathrm{O_2(g)} \rightarrow \mathrm{TiO_2(s)} + 2\mathrm{Cl_2(g)}$ at $1400^\circ\mathrm{C}$, $\Delta H^\circ = -179\ \mathrm{kJ/mol}$. The $\mathrm{Cl_2}$ is recycled. (a) Step 1: A reactor processes $500\ \mathrm{kg/hr}$ of $\mathrm{TiO_2}$ ($M = 79.9$) with stoichiometric $\mathrm{Cl_2}$ and excess C. What mass of $\mathrm{Cl_2}$ ($M = 71.0$) is consumed per hour? (b) What mass of $\mathrm{TiCl_4}$ ($M = 189.7$) is produced per hour at 95% yield? (c) The $\mathrm{TiCl_4}$ is purified by distillation. How much heat is needed per hour to vaporize the $\mathrm{TiCl_4}$? (d) Step 3: The $\mathrm{TiCl_4}$ vapor is burned with $\mathrm{O_2}$ in a flame reactor. The reaction produces $\mathrm{TiO_2}$ particles and regenerates $\mathrm{Cl_2}$. What mass of $\mathrm{Cl_2}$ is regenerated per hour? (The $\mathrm{Cl_2}$ is recycled to Step 1 — what fraction of the Step 1 $\mathrm{Cl_2}$ demand does this meet?) (e) The flame reactor operates at $1400^\circ\mathrm{C}$.

Preheating the O_2 from $25^\circ C$ to $1400^\circ C$ requires heat. $C_{p,O_2} = 1.05\ J\ g^{-1}\ K^{-1}$ (average over the range). For the O_2 needed in (d), what mass of O_2 is required, and how much preheat energy (kJ/hr) is needed? (f) The exothermic oxidation (Step 3) releases $179\ kJ/mol\ TiCl_4$. This heat could theoretically preheat the O_2. Is the reaction exothermic enough to supply the preheat from (e)? Calculate the heat released per hour and compare. (g) The TiO_2 particles are collected. The product specification requires $99.0\%\ TiO_2$ purity. If the crude product contains $97.5\%\ TiO_2$, how much impurity (kg per ton of crude) must be removed? [Cards: 1, 8, 9, 10, 11, 12, 13, 33, 38, 39, 40, 41, 42, 121, 122, 123, 160, 162, 197]

ULTRA-SEMICONDUCTOR — 10 Problems (Level 5, 7–10 domains)

> Semiconductor manufacturing is the most chemically intensive industry per gram of product. A single chip passes through 300+ chemical steps. These problems chain gas delivery, surface kinetics, plasma physics, thermal budgets, and waste treatment into single multi-domain exercises.

C221–C225: Integrated Front-End Processes

C221. [7 domains: Stoich + Gases + Kinetics + Thermo + Equilibrium + Solutions + Bridges] Thermal oxidation of Si with HCl gettering and dopant redistribution. A batch furnace oxidizes 100 wafers ($200\ mm$ diameter) at $1000^\circ C$ in dry O_2 with 3% HCl by volume. (a) The furnace tube volume is $50\ L$. At the process pressure of $1.00\ atm$ and $1000^\circ C$, how many moles of gas are in the tube? ($R = 0.08206$) What are the partial pressures of O_2 and HCl? (b) The target oxide thickness is $50\ nm$ (density $2.20\ g/cm^3$, $M_{SiO_2} = 60.1$). What total mass of SiO_2 grows across all wafers? What mass of Si is consumed? ($M_{Si} = 28.1$) (c) The oxidation follows the linear-parabolic Deal-Grove model. In the linear regime, the rate is $dx/dt = k_L C_{O_2}$, where C_{O_2} is the O_2 concentration at the Si surface. C_{O_2} is governed by Henry's Law: $C = K_H P_{O_2}$ where $K_H = 5.0 \times 10^{-4}\ M/atm$ at $1000^\circ C$. If the linear rate constant $k_L = 2.0 \times 10^{-4}\ cm^2\ mol^{-1}\ s^{-1}$, what is the initial oxidation rate in nm/min? ($1\ M = 10^{-3}\ mol/cm^3$) (d) HCl gettering removes metal impurities by forming volatile metal chlorides. If the incoming O_2 contains $10\ ppb$ Fe (by atoms relative to O_2), and all Fe is converted to $FeCl_3$ (bp $316^\circ C$), what mass of Fe ($M = 55.8$) enters the furnace per batch? (e) The HCl also affects the oxidation rate by forming H_2O: $4HCl + O_2 \rightleftharpoons 2H_2O + 2Cl_2$. At $1000^\circ C$, $K_p = 0.35$. If the initial $P_{HCl} = 0.03\ atm$ and $P_{O_2} = 0.97\ atm$, what is the equilibrium P_{H_2O}? H_2O accelerates oxidation — does this equilibrium produce significant H_2O? [Cards: 1, 8, 9, 10, 19, 26, 27, 28, 76, 78, 80, 89, 93, 95, 158, 160, 186]

C222. [8 domains: Stoich + Gases + Kinetics + Thermo + Equilibrium + Electrochem + Solutions + Bridges] Plasma-enhanced CVD (PECVD) of Si_3N_4 from SiH_4, NH_3, and N_2 in a parallel-plate reactor at $300^\circ C$ and $2.0\ torr$. The plasma dissociates the precursor gases into radicals, dramatically enhancing the deposition rate. (a) The overall reaction is $3SiH_4 + 4NH_3 \rightarrow Si_3N_4 + 12H_2$. The deposition rate is $200\ nm/min$ on a $300\ mm$ wafer (density of $Si_3N_4 = 3.44\ g/cm^3$, $M = 140.3$). What mass of Si_3N_4 is deposited per wafer per minute? (b) The precursor gases are fed at $SiH_4 : 50\ sccm, NH_3 : 200\ sccm, N_2 : 500\ sccm$ ($1\ sccm = 1\ cm^3/min$ at STP). What is the partial pressure of SiH_4 in the reactor (total $P = 2.0\ torr$)? (c) Only 5% of the SiH_4 fed is actually incorporated into the film (the rest is pumped away). What is the SiH_4 utilization efficiency? Verify: from (a), how many mol/min of Si are deposited? From (b), how many mol/min of SiH_4 are fed? (d) The plasma power is $500\ W\ (J/s)$. If 40% of this power goes into dissociating NH_3 ($N-H$ bond energy: $391\ kJ/mol$), how many mol/min of NH_3 can be dissociated? Compare to the NH_3

feed rate from (b) — is the plasma power sufficient? (e) The effluent gas contains unreacted $\mathrm{SiH_4}$, which is pyrophoric (ignites in air). It is treated by a burn box: $\mathrm{SiH_4 + 2O_2 \rightarrow SiO_2 + 2H_2O}$. If 95% of the $\mathrm{SiH_4}$ is unreacted, what mass of $\mathrm{SiO_2}$ ($M = 60.1$) accumulates in the burn box per day? (f) The burn box operates at $800°\mathrm{C}$. The heat from the $\mathrm{SiH_4}$ combustion ($\Delta H^\circ_{\mathrm{comb,SiH_4}} = -1517\ \mathrm{kJ/mol}$) is recovered. How much heat (kJ/day) is recovered from the unreacted $\mathrm{SiH_4}$? (g) The $\mathrm{SiO_2}$ dust from the burn box must be disposed of. If the disposal cost is $\$500$ per metric ton, what is the annual disposal cost? [Cards: 1, 6, 7, 8, 9, 10, 11, 13, 19, 20, 26, 27, 33, 39, 40, 41, 42, 76, 78, 158, 160, 186]

C223. [7 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Solutions + Bridges] Low-pressure CVD (LPCVD) of polysilicon from $\mathrm{SiH_4}$ in a horizontal tube furnace at $620°\mathrm{C}$ and $0.30\ \mathrm{torr}$. The deposition is rate-limited by the surface reaction (first-order in $\mathrm{SiH_4}$ partial pressure at the wafer surface). Gas-phase diffusion through the boundary layer also matters. (a) The furnace tube is $1.5\ \mathrm{m}$ long and $0.20\ \mathrm{m}$ in diameter. 100 wafers ($200\ \mathrm{mm}$ diameter) are spaced $10\ \mathrm{mm}$ apart. What is the free volume in the tube (tube volume minus wafer volume)? (b) $\mathrm{SiH_4}$ is fed at $100\ \mathrm{sccm}$ (STP). The total pressure is $0.30\ \mathrm{torr}$ ($= 4.0 \times 10^{-4}\ \mathrm{atm}$). Using $PV = nRT$, what is the volumetric flow rate at process conditions ($620°\mathrm{C}$, $0.30\ \mathrm{torr}$)? (c) The residence time $\tau = V/Q$ where V is the free volume and Q is the volumetric flow rate at process conditions. What is the residence time in seconds? (d) $\mathrm{SiH_4}$ decomposes by first-order kinetics: Rate = $k[\mathrm{SiH_4}]$ with $k = 0.15\ \mathrm{s^{-1}}$ at $620°\mathrm{C}$ (see C183). What fraction of $\mathrm{SiH_4}$ decomposes during one pass through the furnace? ($f = 1 - e^{-k\tau}$) (e) The deposition rate is proportional to the $\mathrm{SiH_4}$ partial pressure. The $\mathrm{SiH_4}$ is depleted along the tube. If the inlet partial pressure is P_0, the partial pressure at position x is $P(x) = P_0 e^{-kx/v}$ where v is the linear gas velocity. The velocity $v = Q/A_{\mathrm{cross}}$, where A_{cross} is the open cross-sectional area of the tube (tube area minus wafer edge area). Calculate v (cm/s). Then find P/P_0 at $x = 75\ \mathrm{cm}$ (halfway). (f) To compensate for depletion, the furnace temperature is ramped $+10°\mathrm{C}$ from inlet to outlet. Using Arrhenius with $E_a = 160\ \mathrm{kJ/mol}$ (for polysilicon deposition from $\mathrm{SiH_4}$), by what factor does k increase over this $10°\mathrm{C}$ ramp? ($R = 8.314$) Does this compensate for the depletion in (e)? (g) The deposited film thickness uniformity must be within $\pm 3\%$ across all wafers. Based on the depletion profile from (e), is a temperature ramp necessary, or can the process work without it? [Cards: 1, 6, 7, 8, 19, 20, 21, 23, 26, 30, 76, 81, 83, 84, 85, 158, 160, 186, 197]

C224. [7 domains: Stoich + Gases + Thermo + Kinetics + Acid-Base + Equilibrium + Solutions] Post-etch residue removal uses a sequence of wet cleans. After a plasma etch that leaves fluorocarbon polymer residue ($\approx \mathrm{CF_2}$), wafers undergo: (1) $\mathrm{O_2}$ plasma ash to remove polymer; (2) dilute HF dip to remove damaged oxide; (3) SC1 ($\mathrm{NH_4OH/H_2O_2/H_2O}$) to remove particles; (4) SC2 ($\mathrm{HCl/H_2O_2/H_2O}$) to remove metals. (a) The $\mathrm{O_2}$ plasma ash converts the polymer to $\mathrm{CO_2}$ and $\mathrm{SiF_4}$ (from underlying Si etch). If the polymer is $(\mathrm{CF_2})_n$ with $n \approx 1000$, and the total polymer mass per wafer is $0.50\ \mathrm{mg}$, write the idealized ash reaction: $\mathrm{(CF_2)_n + O_2 \rightarrow CO_2 + 2F}$ (F atoms then form $\mathrm{SiF_4}$ with Si from the wafer). What volume of $\mathrm{O_2}$ at STP is consumed per wafer? (b) The HF dip is $100 : 1\ \mathrm{H_2O : HF}$ (by volume) using 49% HF stock. The final solution is 0.50% HF by mass. What is the molarity of HF in the dip? (Density $1.00\ \mathrm{g/mL}$, $M_{\mathrm{HF}} = 20.0$) The dip removes $2.0\ \mathrm{nm}$ of thermally grown $\mathrm{SiO_2}$ (density $2.20\ \mathrm{g/cm^3}$) from a $300\ \mathrm{mm}$ wafer. How many moles of HF are consumed? ($\mathrm{SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O}$) (c) The SC1 solution ($1:1:5\ \mathrm{NH_4OH/H_2O_2/H_2O}$) is prepared at $75°\mathrm{C}$. At this temperature, $\mathrm{H_2O_2}$ decomposes: $\mathrm{2H_2O_2 \rightarrow 2H_2O + O_2}$, first-order with $t_{1/2} = 8.0\ \mathrm{min}$ at $75°\mathrm{C}$. If the initial $[\mathrm{H_2O_2}] = 1.5\ \mathrm{M}$, what is the concentration after $30\ \mathrm{min}$? How often should the bath be recharged? (d) The SC2 solution ($1:1:6\ \mathrm{HCl/H_2O_2/H_2O}$) is at $75°\mathrm{C}$. The high temperature and acidity cause accelerated $\mathrm{H_2O_2}$ decomposition. The decomposition is catalyzed by $\mathrm{H^+}$: Rate = $k[\mathrm{H_2O_2}][\mathrm{H^+}]$ with $k = 1.8 \times 10^{-4}\ \mathrm{M^{-1}\ s^{-1}}$ at $75°\mathrm{C}$. If $[\mathrm{HCl}] = 1.5\ \mathrm{M}$ and initial $[\mathrm{H_2O_2}] = 1.2\ \mathrm{M}$, what is the half-life of $\mathrm{H_2O_2}$ in SC2? Compare to SC1 — why is SC2 much less stable? (e) The waste from all these cleaning steps is combined and neutralized with

$\mathrm{Ca(OH)_2}$ before discharge. The combined waste contains $0.50\ \mathrm{mol}$ HF, $0.20\ \mathrm{mol}$ HCl, and $0.10\ \mathrm{mol}$ $\mathrm{NH_4OH}$ per wafer batch (25 wafers). What mass of $\mathrm{Ca(OH)_2}$ ($M = 74.1$) is needed per batch? (Neutralization: $\mathrm{Ca(OH)_2 + 2H^+ \rightarrow Ca^{2+} + 2H_2O}$; calculate total moles of $\mathrm{H^+}$) [Cards: 1, 8, 9, 10, 11, 19, 33, 76, 81, 83, 86, 101, 107, 108, 109, 133, 134, 158, 160]

C225. [8 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Electrochem + Solutions + Bridges] Ion implantation of arsenic into silicon, followed by rapid thermal annealing (RTA) for dopant activation and damage repair, then electrochemical measurement of sheet resistance. (a) $\mathrm{As^+}$ ions are implanted at $50\ \mathrm{keV}$ with a dose of $5.0 \times 10^{15}\ \mathrm{ions/cm^2}$ into a $300\ \mathrm{mm}$ Si wafer. What is the total number of As atoms implanted? What mass of As ($M = 74.9$) is this? (b) The implanted As atoms are initially mostly interstitial (not electrically active). RTA at $1050^\circ\mathrm{C}$ for $10\ \mathrm{s}$ activates 98% of the dopant. The diffusion coefficient of As in Si at $1050^\circ\mathrm{C}$ follows $D = D_0 e^{-E_a/(RT)}$ with $D_0 = 0.32\ \mathrm{cm^2/s}$ and $E_a = 3.43\ \mathrm{eV}$. ($1\ \mathrm{eV} = 96.5\ \mathrm{kJ/mol}$, $R = 8.314$) Calculate D and the diffusion length $\sqrt{Dt}$. Does the dopant profile broaden significantly during the $10\ \mathrm{s}$ anneal? (c) After activation, the sheet resistance R_s is measured by a four-point probe. $R_s = \rho/t$ where $\rho = 1/(qn\mu)$ is the resistivity. $q = 1.602 \times 10^{-19}\ \mathrm{C}$ (electron charge), n is the carrier concentration (activated As atoms per cm³), and $\mu = 50\ \mathrm{cm^2/V\ s}$ is the electron mobility. The implant depth is approximately $0.10\ \mu\mathrm{m}$. Calculate the carrier concentration n (atoms/cm³), the resistivity ρ, and the sheet resistance R_s in $\Omega/\square$. (d) A separate test structure is a Van der Pauw cross for Hall effect measurement. Applying a current of $1.00\ \mathrm{mA}$ through the sample from (c) and a perpendicular magnetic field of $0.50\ \mathrm{T}$ produces a Hall voltage V_H. The Hall coefficient $R_H = V_H t/(IB) = 1/(nq)$. Calculate R_H and V_H. (e) The RTA chamber is purged with $\mathrm{N_2}$ gas at $1.00\ \mathrm{atm}$. Before annealing, the chamber (volume $5.0\ \mathrm{L}$) must have $\mathrm{O_2}$ reduced below $1.0\ \mathrm{ppm}$. Initially, the chamber is full of air ($21\%\ \mathrm{O_2}$). How many purge cycles (evacuate to $0.01\ \mathrm{atm}$ and refill with $\mathrm{N_2}$ to $1.00\ \mathrm{atm}$) are needed to reach $<$ $1.0\ \mathrm{ppm\ O_2}$? (Each cycle reduces $\mathrm{O_2}$ by the pressure ratio: $P_{\mathrm{O_2,new}} = P_{\mathrm{O_2,old}} \times (0.01/1.00)$.) (f) After RTA, the wafer is rapidly cooled from $1050^\circ\mathrm{C}$ to $100^\circ\mathrm{C}$ in $20\ \mathrm{s}$. The wafer's heat capacity is approximately that of Si: $C_p = 0.705\ \mathrm{J\ g^{-1}\ K^{-1}}$, wafer mass = $128\ \mathrm{g}$ ($300\ \mathrm{mm}$, $0.775\ \mathrm{mm}$ thick, $\rho_{\mathrm{Si}} = 2.33\ \mathrm{g/cm^3}$). Verify the wafer mass. What is the average cooling rate in K/s? How much heat (kJ) must be removed? [Cards: 1, 4, 5, 19, 21, 26, 33, 81, 84, 85, 158, 186, 197, 204, 205]

C226–C230: Integrated Back-End & Advanced Patterning

C226. [7 domains: Stoich + Gases + Thermo + Kinetics + Electrochem + Solutions + Bridges] Copper dual-damascene: (1) Barrier/seed PVD of TaN/Ta and Cu seed; (2) Cu electroplating to fill vias and trenches; (3) CMP to remove excess Cu; (4) Post-CMP clean. A $300\ \mathrm{mm}$ wafer has 10^{10} vias (each $0.050\ \mu\mathrm{m}$ diameter, $0.50\ \mu\mathrm{m}$ deep) and 5×10^9 trenches ($0.050\ \mu\mathrm{m}$ wide, $0.20\ \mu\mathrm{m}$ deep, average length $5.0\ \mu\mathrm{m}$). (a) Calculate the total volume of Cu to fill all features on one wafer. (b) Cu electroplating: $\mathrm{Cu^{2+} + 2e^- \rightarrow Cu}$. The plating current is $10.0\ \mathrm{A}$ for a $300\ \mathrm{mm}$ wafer. Theoretically (at 100% current efficiency), how long (in seconds) to deposit the Cu volume from (a)? ($\rho_{\mathrm{Cu}} = 8.96\ \mathrm{g/cm^3}$, $M_{\mathrm{Cu}} = 63.5$, $F = 96{,}485$) (c) In practice, the plating also deposits a $0.50\ \mu\mathrm{m}$ "overburden" of Cu across the entire wafer surface. What additional Cu volume and plating time does this require? (d) The CMP step removes the overburden and the barrier layer. The CMP slurry contains $5.0\%\ \mathrm{H_2O_2}$ (see C206) and abrasive $\mathrm{SiO_2}$ nanoparticles ($50\ \mathrm{nm}$ diameter, density $2.20\ \mathrm{g/cm^3}$). The slurry has 1.0×10^{14} particles per mL. What is the mass percent of $\mathrm{SiO_2}$ in the slurry? (e) After CMP, the wafer is cleaned with a solution of 0.50% HF + 1.0% citric acid ($\mathrm{C_6H_8O_7}$, a weak triprotic acid, $\mathrm{p}K_{a1} = 3.13$) by mass. What is the pH of the citric acid component? (Assume only the first dissociation matters, $M_{\mathrm{citric}} = 192.1$, density $1.00\ \mathrm{g/mL}$.) (f) The CMP waste slurry contains $100\ \mathrm{mg/L}$ Cu. Before discharge, Cu must be below $1.0\ \mathrm{mg/L}$. The waste ($1000\ \mathrm{L/day}$) is treated by electroplating the Cu onto a sacrificial cathode at $50\ \mathrm{A}$. How many hours per day must the treatment cell operate? ($F = 96{,}485$) [Cards: 1, 4, 5, 8, 9, 10, 19, 76, 101, 104, 109, 121, 122, 129, 131, 132, 133, 134, 135, 141, 142, 158, 160]

C227. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Bridges] Atomic Layer Deposition (ALD) of $\mathrm{HfO_2}$ high-κ dielectric using $\mathrm{HfCl_4}$ and $\mathrm{H_2O}$ precursors in a cross-flow reactor at $300^\circ\mathrm{C}$ and $0.50\ \mathrm{torr}$. ALD is a surface-saturated, self-limiting process: each precursor pulse deposits exactly one sub-monolayer. (a) The ALD half-reactions: Pulse A: $\mathrm{Si{-}OH_{(s)} + HfCl_4(g) \rightarrow Si{-}O{-}HfCl_3(s) + HCl(g)}$. Pulse B: $\mathrm{HfCl_{(s)} + H_2O(g) \rightarrow Hf{-}OH_{(s)} + HCl(g)}$. The net reaction per cycle is $\mathrm{HfCl_4 + 2H_2O \rightarrow HfO_2 + 4HCl}$. Each cycle deposits $0.12\ \mathrm{nm}$ of $\mathrm{HfO_2}$ (density $9.68\ \mathrm{g/cm^3}$, $M = 210.5$). For a $300\ \mathrm{mm}$ wafer, how many Hf atoms are deposited per cm² per cycle? ($N_A = 6.022 \times 10^{23}$) (b) A typical ALD cycle: $\mathrm{HfCl_4}$ pulse $0.50\ \mathrm{s}$, $\mathrm{N_2}$ purge $5.0\ \mathrm{s}$, $\mathrm{H_2O}$ pulse $0.30\ \mathrm{s}$, $\mathrm{N_2}$ purge $5.0\ \mathrm{s}$. What is the cycle time? To deposit a $3.0\ \mathrm{nm}$ film, how many cycles and what total process time? (c) The $\mathrm{HfCl_4}$ precursor is a solid at room temperature ($P_{\mathrm{vap}} = 0.10\ \mathrm{torr}$ at $150^\circ\mathrm{C}$). The precursor bottle is heated to $150^\circ\mathrm{C}$. The vapor is transported to the reactor by $50\ \mathrm{sccm}$ of $\mathrm{N_2}$ carrier gas. Using Dalton's Law, what is the molar flow rate of $\mathrm{HfCl_4}$? (Total P at the bottle = $1.0\ \mathrm{atm}$; the carrier gas saturates with $\mathrm{HfCl_4}$ vapor. $P_{\mathrm{HfCl_4}}/P_{\mathrm{total}} = n_{\mathrm{HfCl_4}}/n_{\mathrm{total}}$.) (d) The $\mathrm{HfCl_4}$ utilization (fraction of precursor that reacts on the wafer) is only 2.0%. How much $\mathrm{HfCl_4}$ ($M = 320.3$) is consumed per wafer (wasted + used) for the $3.0\ \mathrm{nm}$ film? (e) The unreacted $\mathrm{HfCl_4}$ and HCl byproduct are trapped in a scrubber containing $\mathrm{CaCO_3}$: $\mathrm{2HCl + CaCO_3 \rightarrow CaCl_2 + CO_2 + H_2O}$. How many grams of $\mathrm{CaCO_3}$ ($M = 100.1$) are consumed per 100 wafers? ($\mathrm{HfCl_4}$ hydrolyzes in the scrubber: $\mathrm{HfCl_4 + 2H_2O \rightarrow HfO_2 + 4HCl}$, so total HCl = from precursor + from reaction.) (f) To improve throughput, the $\mathrm{N_2}$ purge time is reduced from 5.0 to $2.0\ \mathrm{s}$. However, if CVD-like reactions occur in the gas phase (from inadequate purging), particles form. The particle formation rate follows Arrhenius: $E_a = 85\ \mathrm{kJ/mol}$, and is first-order in residual precursor concentration. If purging follows first-order decay with time constant $\tau = 1.0\ \mathrm{s}$, the residual concentration after $5.0\ \mathrm{s}$ is $C_0 e^{-5}$. After $2.0\ \mathrm{s}$, it is $C_0 e^{-2}$. By what factor does the particle formation rate increase when reducing purge from 5.0 to $2.0\ \mathrm{s}$? [Cards: 1, 8, 9, 10, 19, 26, 27, 76, 81, 83, 84, 85, 133, 158, 160, 186]

C228. [8 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Acid-Base + Bridges] Extreme ultraviolet (EUV) lithography uses $13.5\ \mathrm{nm}$ photons generated by a laser-produced plasma from Sn droplets. (a) The EUV source vaporizes Sn ($M = 118.7$) droplets. A droplet of $30\ \mu\mathrm{m}$ diameter ($\rho_{\mathrm{Sn}} = 7.29\ \mathrm{g/cm^3}$) is hit by a laser. How many Sn atoms are in one droplet? (b) The Sn plasma produces EUV light by $10\times$ ionizing Sn. The ionization energy to reach $\mathrm{Sn^{10+}}$ from neutral Sn is approximately the sum of the first 10 ionization energies: IE_1 = 709, IE_2 = 1412, IE_3 = 2943, IE_4 = 3930, IE_5 = 7460, IE_6 = 8600, IE_7 = 9900, IE_8 = 11200, IE_9 = 12500, IE_{10} = 13800 kJ/mol. What is the total energy (kJ/mol) to produce $\mathrm{Sn^{10+}}$? What fraction of the laser pulse energy ($200\ \mathrm{mJ}$ per droplet) does this represent? ($N_A = 6.022 \times 10^{23}$) (c) Sn debris coats the collector mirror. The mirror is cleaned by hydrogen radical etching: $\mathrm{Sn(s) + 4H\cdot(g) \rightarrow SnH_4(g)}$. $\mathrm{H\cdot}$ is generated by flowing $\mathrm{H_2}$ through a hot filament ($2000^\circ\mathrm{C}$). The equilibrium $\mathrm{H_2 \rightleftharpoons 2H}$ has $K_p = 1.2 \times 10^{-3}$ at $2000^\circ\mathrm{C}$. If $P_{\mathrm{H_2}} = 0.10\ \mathrm{atm}$ at the filament, what is P_{H}? What fraction of $\mathrm{H_2}$ is dissociated? (d) The EUV photoresist is a chemically amplified resist (CAR). Upon exposure, a photoacid generator (PAG) produces $\mathrm{H^+}$: PAG + $h\nu$ → $\mathrm{H^+}$ + byproducts. The quantum yield is 0.30 (0.30 $\mathrm{H^+}$ per photon). The EUV dose is $20\ \mathrm{mJ/cm^2}$. The photon energy $E = hc/\lambda$ where $h = 6.626 \times 10^{-34}\ \mathrm{J\ s}$, $c = 3.00 \times 10^8\ \mathrm{m/s}$, $\lambda = 13.5 \times 10^{-9}\ \mathrm{m}$. (e) Calculate the number of photons per cm² in the dose. Then calculate $[\mathrm{H^+}]$ generated in the resist (thickness $50\ \mathrm{nm}$, density $1.0\ \mathrm{g/cm^3}$). The resist contains 5.0% PAG by mass ($M_{\mathrm{PAG}} = 500$). (f) During post-exposure bake (PEB) at $110^\circ\mathrm{C}$, the $\mathrm{H^+}$ catalyzes a deprotection reaction that switches the resist solubility. The deprotection is first-order in $[\mathrm{H^+}]$ with $k = 2.0 \times 10^3\ \mathrm{M^{-1}\ s^{-1}}$ at $110^\circ\mathrm{C}$. The PEB time is $60\ \mathrm{s}$. What fraction of the protecting groups is removed? (Assume pseudo-first-order: $k' = k[\mathrm{H^+}]$, fraction = $1 - e^{-k't}$.) (g) The exposed resist is developed in 2.38% TMAH (see C204). After development, the pattern quality is assessed. The line-width roughness (LWR) is $3.0\ \mathrm{nm}\ (3\sigma)$. If the target critical dimension (CD) is $20\ \mathrm{nm}$, what is the LWR as a percentage of CD? [Cards: 1, 4, 5, 19, 20, 26, 33, 39, 53, 60, 76, 81, 83, 89, 93, 95, 101, 104, 107, 109, 133, 135, 137, 158]

C229. [7 domains: Stoich + Gases + Thermo + Kinetics + Electrochem + Solutions + Bridges] Chemical vapor deposition of tungsten (W) for contact plugs using $\mathrm{WF_6}$ and $\mathrm{H_2}$ in a cold-wall reactor at $400°\mathrm{C}$, $30\ \mathrm{torr}$. Tungsten CVD is used to fill high-aspect-ratio contacts because of its excellent conformality. (a) The overall reaction is $\mathrm{WF_6(g) + 3H_2(g) \rightarrow W(s) + 6HF(g)}$, $\Delta H° = +69\ \mathrm{kJ/mol}$ (actually endothermic for this specific reduction; the Si reduction is exothermic). What does the positive $\Delta H°$ imply about temperature control? Why must the reactor be heated despite the reaction being endothermic? (b) A $300\ \mathrm{mm}$ wafer has 5.0×10^9 contacts (each $0.10\ \mu\mathrm{m}$ diameter, $1.0\ \mu\mathrm{m}$ deep). What total volume and mass of W ($\rho = 19.3\ \mathrm{g/cm^3}$, $M = 183.8$) is needed? (c) The deposition rate is limited by $\mathrm{WF_6}$ supply at low pressures. The feed gas is $10\%\ \mathrm{WF_6}$, $30\%\ \mathrm{H_2}$, 60% Ar at $30\ \mathrm{torr}$ total. What is $P_{\mathrm{WF_6}}$? (d) The sticking coefficient of $\mathrm{WF_6}$ on W at $400°\mathrm{C}$ is $\gamma = 0.10$ (probability that an impinging molecule reacts). The impingement rate (molecules per cm² per second) from kinetic theory is $Z = \dfrac{P}{\sqrt{2\pi MRT}}$, with P in Pa ($1\ \mathrm{atm} = 101{,}325\ \mathrm{Pa}$), M in kg/mol ($M_{\mathrm{WF_6}} = 0.2978\ \mathrm{kg/mol}$), $R = 8.314$, T in K. Calculate Z and the deposition rate in nm/s. ($\rho_{\mathrm{W}} = 19.3\ \mathrm{g/cm^3}$; convert molecules/cm²/s to g/cm²/s to nm/s.) (e) The byproduct HF etches $\mathrm{SiO_2}$: $\mathrm{SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O}$. The reactor wall has a quartz ($\mathrm{SiO_2}$) liner. If $5.0\ \mathrm{g}$ of W are deposited per wafer, how much HF is produced? What mass of the quartz liner is etched per wafer? ($M_{\mathrm{SiO_2}} = 60.1$) (f) The effluent HF must be scrubbed. A wet scrubber uses NaOH: $\mathrm{HF + NaOH \rightarrow NaF + H_2O}$. If the fab runs 1000 wafers per day through this process, what mass of NaOH ($M = 40.0$) is consumed daily? (g) Excess $\mathrm{WF_6}$ is also scrubbed: $\mathrm{WF_6 + 8NaOH \rightarrow Na_2WO_4 + 6NaF + 4H_2O}$. Only 15% of the $\mathrm{WF_6}$ is consumed in the reactor (the rest is pumped out). What additional NaOH is needed per day? Total NaOH consumption? [Cards: 1, 8, 9, 10, 11, 19, 20, 26, 27, 30, 33, 39, 40, 41, 76, 78, 84, 101, 107, 108, 129, 131, 133, 158, 160, 197, 199]

C230. [10 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Electrochem + Acid-Base + Solutions + Bridges + Unit Conversions] End-to-end chip manufacturing environmental footprint. A logic fab processes $50{,}000$ wafers ($300\ \mathrm{mm}$) per month. Calculate the annual environmental metrics using the full vocabulary deck. (a) **[Stoich + Gases]** The fab's $\mathrm{N_2}$ consumption (for purging) is $500\ \mathrm{m^3}$ per wafer (measured at STP). What is the annual $\mathrm{N_2}$ consumption in metric tons? ($M_{\mathrm{N_2}} = 28.0$, $1\ \mathrm{m^3} = 1000\ \mathrm{L}$) (b) **[Gases + Thermo]** The $\mathrm{N_2}$ is generated on-site by cryogenic air separation. Air is cooled from $25°\mathrm{C}$ to $-196°\mathrm{C}\ (77\ \mathrm{K})$. Air is $78\%\ \mathrm{N_2}$ by volume, $C_{p,\mathrm{air}} = 1.01\ \mathrm{J\ g^{-1}\ K^{-1}}$, $M_{\mathrm{air}} \approx 29.0$. What is the theoretical minimum energy (GJ/yr) to cool the air needed for the annual $\mathrm{N_2}$ production? (Ignore latent heats of $\mathrm{O_2}$ and Ar condensation for this estimate.) (c) **[Electrochem + Stoich]** The fab's ultrapure water (UPW) consumption is $2000\ \mathrm{L}$ per wafer. UPW is produced by reverse osmosis followed by electrodeionization (EDI). EDI uses $0.50\ \mathrm{kWh/m^3}$ of electricity. The electricity is generated from natural gas in a combined-cycle plant at 55% efficiency. Natural gas ($\approx \mathrm{CH_4}$) combustion: $\mathrm{CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O}$, $\Delta H° = -890\ \mathrm{kJ/mol}$. How many metric tons of $\mathrm{CO_2}$ ($M = 44.0$) are emitted annually just from the EDI electricity? ($1\ \mathrm{kWh} = 3.6 \times 10^6\ \mathrm{J}$) (d) **[Acid-Base + Solutions + Stoich]** The fab uses $100\ \mathrm{L}$ of 49% HF (density $1.18\ \mathrm{g/mL}$) per 1000 wafers for etching and cleaning. All fluoride must be precipitated as $\mathrm{CaF_2}$ ($K_{sp} = 3.9 \times 10^{-11}$) before discharge. The waste treatment adds $\mathrm{Ca(OH)_2}$: $\mathrm{2HF + Ca(OH)_2 \rightarrow CaF_2 + 2H_2O}$. What is the annual consumption of $\mathrm{Ca(OH)_2}$ ($M = 74.1$) in metric tons? The discharge limit is $15\ \mathrm{mg/L}$ fluoride. If the total wastewater flow is $5000\ \mathrm{m^3}$ per day, does $\mathrm{CaF_2}$ precipitation achieve the limit? (Calculate $[\mathrm{F^-}]$ in equilibrium with $\mathrm{CaF_2}$, assuming excess $\mathrm{Ca^{2+}}$ at $0.010\ \mathrm{M}$.) (e) **[Thermo + Gases + Kinetics]** The fab's cleanroom HVAC circulates $10^7\ \mathrm{m^3/hr}$ of air. The air is filtered and conditioned from outside conditions ($35°\mathrm{C}$, 80% RH) to cleanroom conditions ($22°\mathrm{C}$, 45% RH). The enthalpy change is approximately $40\ \mathrm{kJ}$ per kg of dry air. Air density at $22°\mathrm{C}$ is $1.20\ \mathrm{kg/m^3}$. What is the annual HVAC energy consumption in GJ? If this energy comes from the same natural gas plant as (c), what are the associated $\mathrm{CO_2}$ emissions? (f) **[Equilibrium + Thermo + Solutions]** The fab's $\mathrm{CO_2}$ footprint from (a-e) totals to X metric tons/yr. A post-combustion amine capture plant is installed that captures 90% of the $\mathrm{CO_2}$ from the natural gas flue. The capture solvent (MEA, monoethanolamine, $\mathrm{HOCH_2CH_2NH_2}$) has a $\mathrm{CO_2}$ loading capacity of $0.50\ \mathrm{mol\ CO_2}$ per mol MEA. MEA's $M = 61.1$, density $1.01\ \mathrm{g/mL}$. What volume of MEA circulates through the capture plant annually? (Assume MEA is regenerated and recycled, volume = one-time fill.) (g) **[Unit Conversions]** Summarize: Express the fab's annual $\mathrm{CO_2}$ emissions per wafer and per cm² of silicon produced (wafer area = $\pi \times (15\ \mathrm{cm})^2$). [Cards: 1, 6, 7, 8, 9, 10, 11, 17, 19, 20, 26, 27, 33, 34, 38, 39, 40, 41, 42, 48, 51, 89, 93, 95, 101, 103, 104, 107, 108, 121, 122, 129, 131, 132, 133, 134, 135, 136, 137, 141, 142, 143, 144, 158, 160, 162, 186, 197, 198, 199, 201, 202, 203]

> **Next step:** Test yourself. Pick one problem from each section. Solve using the method in `chemistry-without-intuition.md`. Check your answers against `chemistry-answers.md`. Repeat until every card reference is fluent.